

Synchronous Ensembles of Hippocampal CA1 Pyramidal Neurons Associated with Theta but not Ripple Oscillations During Novel Exploration

Reviewed Preprint

v2 • February 10, 2025

Revised by authors

Reviewed Preprint

v1 • April 26, 2024

En-Li Chen, Tsai-Wen Chen, Eric R Schreiter, Bei-Jung Lin 

Institute of Neuroscience, National Yang Ming Chiao Tung University, Taipei, Taiwan • Brain Research Center, National Yang Ming Chiao Tung University, Taipei, Taiwan • Janelia Research Campus, Howard Hughes Medical Institute, Ashburn, United States

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eLife Assessment

In this **useful** study, the authors perform voltage imaging of CA1 pyramidal cells in head-fixed mice running on a track while local field potentials (LFPs) are recorded. The authors conclude that synchronous ensembles of neurons are differentially associated with different types of LFP patterns, namely theta and ripples. However, evidence for the claims remains **incomplete**, due to caveats of the experimental approach that were not acknowledged and strong claims that are based on a sparse data set.

<https://doi.org/10.7554/eLife.96718.2.sa4>

Abstract

Synchronous neuronal ensembles play a pivotal role in the consolidation of long-term memory in the hippocampus. However, their organization during the acquisition of spatial memory remains less clear. In this study, we used neuronal population voltage imaging to investigate the synchronization patterns of CA1 pyramidal neuronal ensembles during the exploration of a new environment, a critical phase for spatial memory acquisition. We found synchronous ensembles comprising approximately 40% of CA1 pyramidal neurons, firing simultaneously in brief windows (~25ms) during immobility and locomotion in novel exploration. Notably, these synchronous ensembles were not associated with ripple oscillations but were instead phase-locked to local field potential theta waves. Specifically, the subthreshold membrane potentials of neurons exhibited coherent theta oscillations with a depolarizing peak at the moment of synchrony. Among newly formed place cells, pairs with more robust synchronization during locomotion displayed more distinct place-specific activities. These findings underscore the role of synchronous ensembles in coordinating place cells of different place fields.

Introduction

The establishment of long-term memory involves the initial acquisition and subsequent consolidation of memory traces, processes intricately connected to the modification of synaptic transmission^{1,2}. Synchronous ensembles, characterized by the simultaneous firing of multiple neurons over tens of milliseconds (referred to as population synchrony), play a pivotal role in influencing synaptic plasticity^{3–5}. The hippocampus, a crucial region for long-term memory formation^{6,7}, has been a focal point of synchronous ensemble studies. Within the hippocampus, synchronous ensembles are hypothesized to play roles in reactivating and consolidating labile memory traces⁸. However, their organization during memory acquisition remains less understood.

In the hippocampus, synchronous ensembles among CA1 pyramidal cells (CA1PCs) are predominantly observed during sharp-wave ripples, the brain waves associated with memory consolidation^{8–13}. During these events, 10–18% of CA1PCs generate spikes, particularly during offline consummatory behavioral states or sleep^{14,15}. These synchronous ensembles are biased toward experience-dependent reactivation, consolidating labile memory traces^{16,17}. Furthermore, synchronous ensembles firing action potentials in short windows of 10–30 milliseconds enhance information processing⁶ and discriminate distinct behavioral contingencies¹⁸. Although synchronous ensembles of CA1PCs during offline memory consolidation are well-characterized, their organization during online acquisition of spatial memory, such as when animals enter a new environment and new place cells form, remains unclear.

Previous studies have identified correlated activities among place cells during theta oscillation, where 1–3% of place cells' spikes co-activate in a forward order matching animals' trajectory in the theta cycles when animals' locations overlap in the place fields of these cells^{19–21}. Because these place cells fire at distinct theta phases in the same cycle, this coordinated firing operates on the timescale of theta periods (~125ms)^{22,23}. Population synchrony on shorter timescales (10–30 ms) is speculated to be reduced by the presence of theta oscillations²².

The subthreshold membrane voltage (subVm), influenced by synaptic inputs and intrinsic conductances, directly triggers spiking activity^{24,25}. Additionally, properties of subVm support the organization of neuronal representation²⁶. Although correlated membrane potentials have been proposed to underlie the synchrony of neuronal ensembles^{27,28}, multicellular recordings of subVms have not yet provided direct experimental evidence. Given the intricate interaction among subVm, spiking, and place cell formation, it is crucial to investigate the temporal relationships of sub- and suprathreshold neuronal activities among CA1PCs during novel exploration.

In this study, we used voltage imaging to investigate the temporal dynamics of sub- and suprathreshold neuronal activities in CA1PCs during novel exploration. We found synchronous ensembles involving approximately 40% of CA1PCs exhibited transient population synchrony and correlated subthreshold membrane potentials when mice ran and stopped in a new maze. Surprisingly, these synchronous ensembles occurred outside ripple episodes and were phase-locked to extracellular and intracellular theta oscillations. Notably, cell pairs with stronger synchronous activities established more distinct place fields that overlapped less. In essence, synchronous ensembles facilitated temporal association among place cells representing diverse features of spatial memory. This plastic network, driven by synchronous ensembles, may contribute to the stabilization of place cells, establishing place fields that effectively tile the environment.

Results

To investigate the temporal relationships between neuronal activities during novel exploration, we simultaneously recorded many neurons as animals explored a new environment for the first time. To achieve this, we employed voltage imaging in conjunction with an air-lifted plastic track that allowed mice to explore an environment while remaining head-fixed under a microscope (**Figure 1A**). To ensure a genuinely novel experience, all pre-training and habituation were performed in a separate room distinct from the one where the imaging experiments were conducted. On the day of imaging, the mice were introduced to the recording room for the first time and initially confined to a specific corner of the track (**Figure 1A** top view). Subsequently, as the imaging session commenced, the confinement was lifted, granting mice the freedom to explore the track while imaging was conducted.

In novel exploration, mice moved at a speed of 8.3 ± 0.9 cm/s during locomotion (speed > 3 cm/s) and spent $53 \pm 7\%$ and $36 \pm 6\%$ of their time in locomotion and immobility (speed < 1 cm/s), respectively (n = 5 mice). Meanwhile, they completed 10 ± 2.5 laps during the 180-s imaging periods (n = 5 mice).

For voltage imaging, we expressed the voltage indicator Voltron2²⁹ in CA1 pyramidal cells (CA1PCs) that were born on embryonic day (E) 14.5, allowing us to measure both suprathreshold and subthreshold membrane potentials from multiple CA1PCs (14 ± 4 cells imaged per field of view (mean $\pm$ s.d.) and, in total, 71 cells imaged from 5 fields of view in 5 mice; **Figure 1B**, **Supplementary Figure 1A, 1B**). Simultaneously, we monitored the animals' positions and speed within the track while recording the local field potential (LFP) from the contralateral side of the hippocampus. Consistent with previous studies, neurons labeled on E14.5 located more on the deep side of the pyramidal layer than those labeled on E17.5 ($t_{(601)} = 22.8$, $p < 0.0001$, Student's *t*-test; **Supplementary Figure 1C, D**).

The mean firing rates of cells ranged from 2.3 to 4.3 Hz (25th-75th percentiles) with a median of 3.2 Hz (n = 71 cells, **Supplementary Figure 2A**). Additionally, most CA1PCs exhibited elevated mean firing rates during immobility relative to locomotion, and their firing rates showed a negative correlation with the animals' speed (**Supplementary Figure 2B, 2C**), consistent with previous research.

Synchronous ensembles among many CA1 pyramidal neurons

Figure 1C shows the fluorescence traces of all neurons in a session as an example. Instances of synchronous activity, where many neurons emitted spikes within a narrow time window, were frequently observed (**Figure 1C**). To investigate synchronous activity within the neuronal population, we selected a reference cell and counted the number of spikes across all other cells that were aligned with the spikes of the reference cell. This approach yielded a grand average cross-correlogram (CCG) capable of elucidating the timing of spikes from the reference cell relative to the combined spikes of the rest of the population. Notably, the grand average CCG exhibited heightened spike counts around zero lags, unlike flattened counts calculated from randomly selected cells of different sessions (**Figure 1D** and **1E**). The full width at half maximum (FWHM) of the peak in the grand average CCG measured 23 ± 14 ms (mean $\pm$ s.d., n = 71 cells from 5 sessions; **Figure 1F**), indicating the timescale of the population synchrony. Furthermore, pairwise CCGs of many cell pairs showed significant peaks compared to jittered data (percentage of significant pairs: 63% (median) n = 497 pairs, **Figure 1-figure supplement 1A** and **1B**). Among pairs with significant peaks, the absolute values of the peak lags ranged from 0.8 to 4.3 ms (25th-75th percentiles) with a median of 2.1 ms, and the FWHM of the peaks ranged from 20 to 46

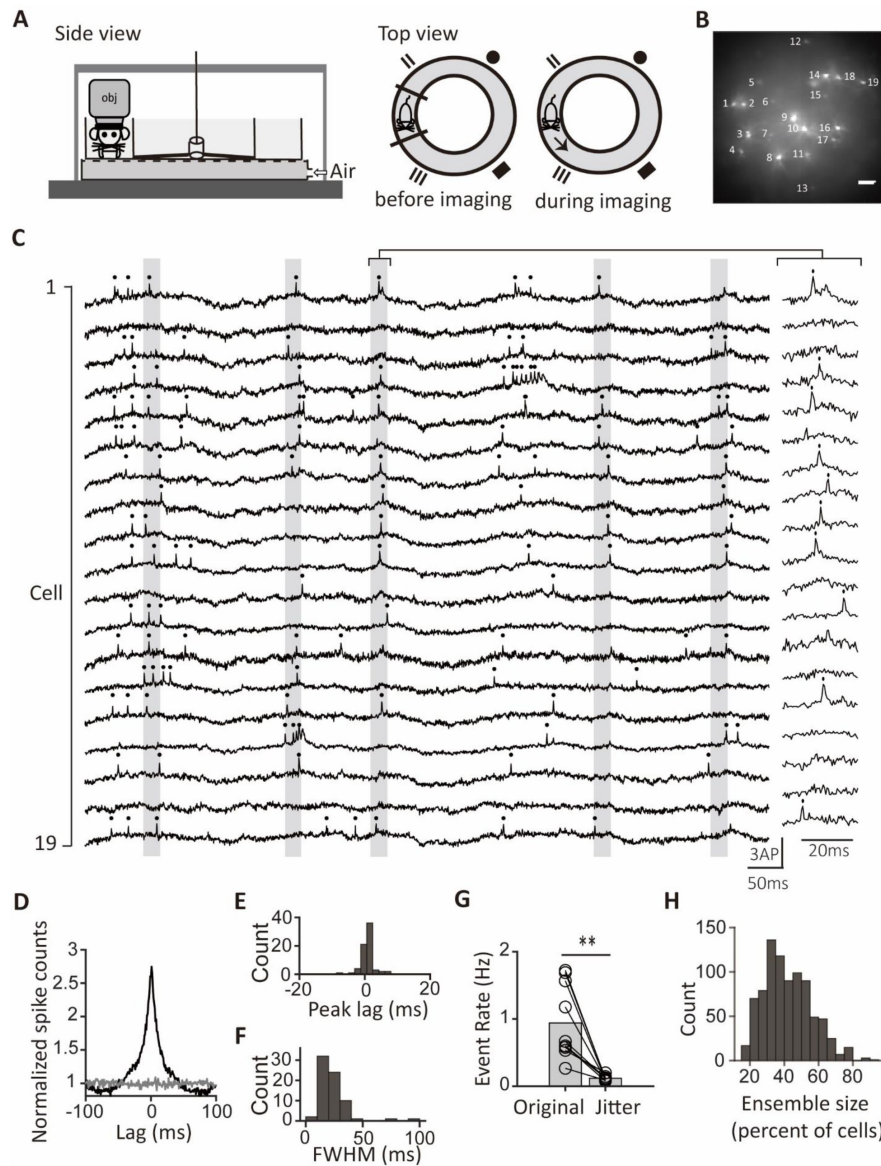


Figure 1.

Population synchrony during novel exploration.

- (A) Schematic of the behavioral setup for voltage imaging. Mice explored a novel one-way track while fluorescence images were captured at the CA1 pyramidal layer. Left: side view; right: top view of the setup.
- (B) Example time-averaged image of CA1 pyramidal cells expressing Voltron2-ST labeled with JF552⁺ HaloTag ligand. Scale bar, 100 μ m.
- (C) Fluorescence traces of cells in an example session with identified synchronous ensembles (labeled with gray vertical bars). Right: an example of magnified fluorescence traces within the bracket, pinpointing the occurrence of the identified synchronous ensemble. Spikes are indicated by dots on the top.
- (D) The grand average cross-correlogram (CCG) averaged across all 71 cells. The gray line represents the mean grand average CCG between reference cells and randomly selected cells from different sessions.
- (E) Distribution of the grand average CCG peak lags of all cells ($n=71$).
- (F) Distribution of the full width at half maximum (FWHM) of the grand average CCGs ($n=71$ cells).
- (G) Pairwise comparison of the event rates of population synchrony between original and jittered data. Bar heights indicate group means. ** $p<0.01$
- (H) Histogram of the ensemble sizes as percentages of cells participating in the synchronous ensembles.

ms (25th-75th percentiles) with a median of 30 ms ($n=315$ pairs, **Figure 1-figure supplement 1C** and **1D**). These results highlight the synchronous spiking between neurons on the millisecond timescale.

To systematically reveal individual synchronous events, we counted the number of spikes from all simultaneously recorded neurons within sliding windows of 25 ms. To compare the summed spikes with controls, we randomly jittered the spike trains to perturb spike timings while maintaining spike rates. We identified moments of population synchrony when spike sums exceeded the average of the jittered spike sums plus four standard deviations (**Figure 1C**). Using the same criterion for event detection, we observed over a 7.8-fold increase in synchronous events in the original data compared to jittered controls (mean event rate: 0.94 ± 0.17 Hz for the original data, 0.12 ± 0.01 Hz for the jittered data, $n=10$ segments, $p=0.002$, $W=55$, Wilcoxon signed-rank test; **Figure 1G**). The observed population synchrony was not attributable to spikes in complex bursts, as synchronous event rates did not differ significantly with or without the inclusion of later spikes in bursts (**Supplementary Figure 3**). Furthermore, we calculated the percentage of neurons participating in these synchronous events to examine the synchronous ensemble sizes. The ensemble sizes ranged from 32% to 53% (25th-75th percentiles) with a median of 42%, which were larger than the synchronous ensemble sizes associated with ripples¹⁴ (**Figure 1H**). These results demonstrate synchronous ensembles of CA1PCs engaging many neurons.

Synchronous ensembles during locomotion and immobility of novel exploration

The hippocampus exhibits state-dependent activities when rodents are engaged in locomotion and immobility^{35,36}. In novel exploration, CA1PCs showed synchronous spiking during locomotion and immobility (**Figure 2A**). To investigate whether these synchronous activities differed between these behavioral states, we compared grant average CCGs and synchronous ensembles under these conditions. During both immobility and locomotion, grant average CCGs showed prominent peaks compared to jittered data (**Figure 2B**). Moreover, synchronous ensembles were rich in both immobility and locomotion periods (**Figure 2A, 2B** and **2D**). Nevertheless, the FWHMs of grant average CCGs were broader in the immobility group compared to the locomotion group (**Figure 2B** and **2C**). Synchronous events occurred more frequently during immobility compared to locomotion (**Figure 2D**). Additionally, ensemble sizes were significantly larger in the immobility group compared to the locomotion group (**Figure 2E**). Taken together, with differences in sizes and kinetics, there were synchronous ensembles during both immobility and locomotion periods of novel exploration.

Synchronous ensembles occur outside ripple episodes

CA1PCs exhibit synchronous activities when the hippocampal network emits ripple waves^{8,14}. To test whether there is a co-occurrence of synchronous ensembles and ripples, we first detected ripples from LFP recordings. Ripples were detected in three of the five recording sessions, and all occurred during immobility (mean ripple rate: 0.04 ± 0.02 Hz during immobility, 0 Hz during locomotion). In the two recording sessions where we did not detect any ripples, we verified the recording quality of the LFP signal by running multiple tests after the recording sessions. Ripples were identified in the same animals during quality tests, indicating that the absence of ripples during novel immobility in the recording session was not due to the deterioration of LFP recordings (**Figure 3-figure supplement 1**).

To visualize both ripples and synchronous ensembles, we plotted voltage traces of neurons alongside simultaneously recorded ripple oscillations. It was typical for synchronous ensembles to occur far away from ripple episodes (**Figure 3A**). To delve into the relationship between neuronal activities and ripples, we aligned traces of ripple power with population synchrony and ripples. In a recording session where 64 synchronous ensembles and 12 ripples were detected

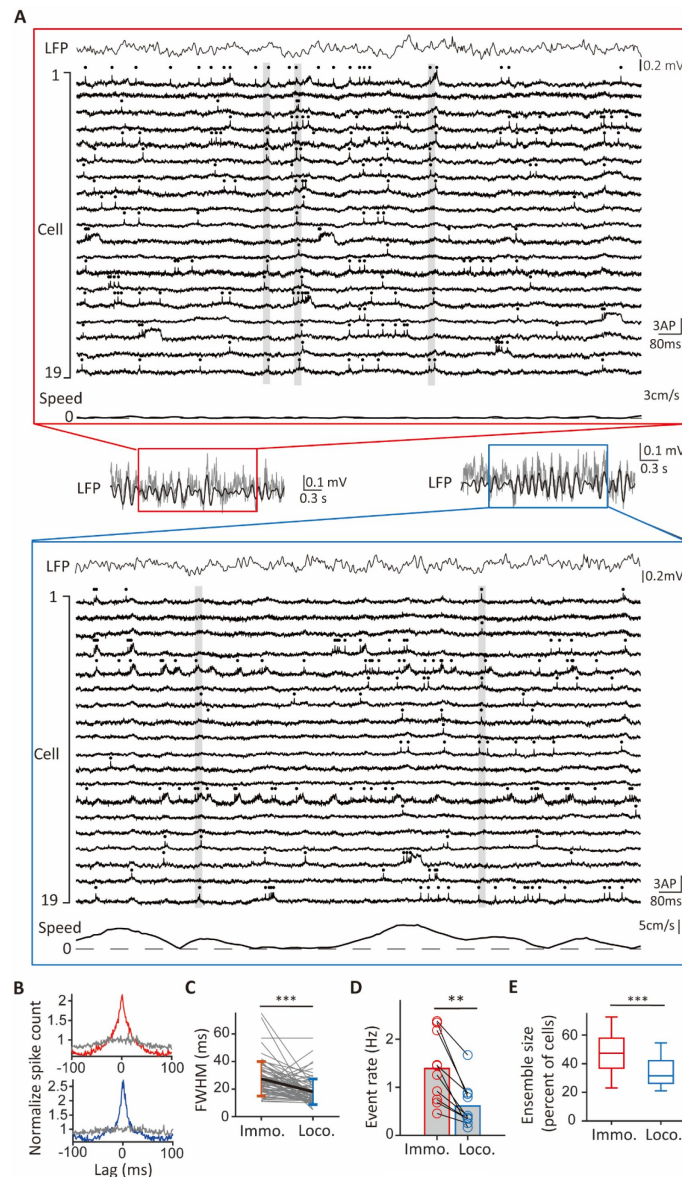


Figure 2.

Comparison of synchronous ensembles between immobility and locomotion.

(A) Example traces of cells in a session with identified synchronous ensembles (gray vertical bars) during periods of immobility (red rectangular frame) and locomotion (blue rectangular frame). Spikes are indicated by dots at the top of individual traces. Top: the simultaneously recorded LFP trace. Bottom: simultaneously recorded speed of the animal. Inset: the band-pass filtered LFP trace at the theta frequency range (black line) overlaid on the raw LFP trace (gray line).

(B) The grand average cross-correlograms (CCGs) during immobility (red) and locomotion (blue) averaged across all cells ($n=71$). The gray lines represent the mean grand average CCGs calculated from jittered data.

(C) Pairwise comparison of peak widths (FWHM) in the grand average CCGs between immobility (red) and locomotion (blue). Vertical bars represent means and standard deviations for both groups (FWHM: 27 ± 12 ms for immobility, 18 ± 9 ms for locomotion, $\text{mean} \pm \text{s.d.}$, $t_{(70)}=5.45$, $p<0.001$, paired Student's t -test, $n=71$ cells). $***p<0.001$

(D) Pairwise comparison of event rates of population synchrony during immobility and locomotion. Bar heights indicate group means (immobility: 1.4 ± 0.7 Hz, locomotion: 0.6 ± 0.5 Hz, $\text{mean} \pm \text{s.d.}$, $n=10$ segments, $W=55$, $p=0.002$, Wilcoxon signed-rank test). $**p<0.01$

(E) Boxplot of ensemble sizes for synchronous ensembles occurring during immobility (red) and locomotion (blue) (median ensemble size: 47% for immobility, $n=446$ events, 32% for locomotion, $n=313$ events, $t_{(757)}=13.54$, $p<0.001$, Student's t -test). $***p<0.001$

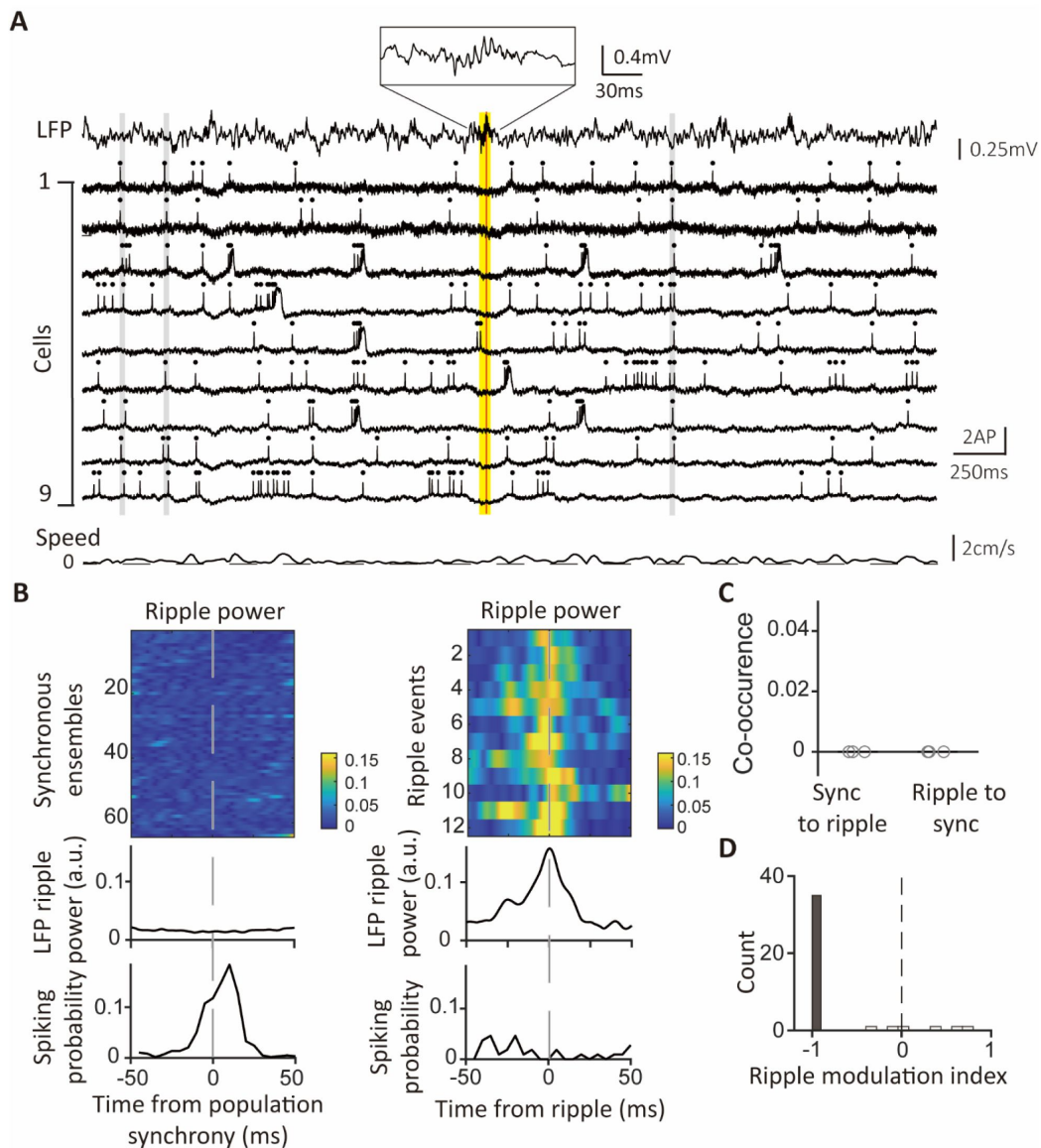


Figure 3.

Synchronous ensembles occur outside ripple episodes.

(A) Example traces with ripples and synchronous ensembles. Top: simultaneously recorded LFP displaying an episode of ripple oscillation. A red vertical line indicates the timing of the peak ripple power in the LFP trace. A yellow vertical bar marks the duration of the ripple episode. Inset at the top: a magnified view of the ripple oscillation. Middle: fluorescence traces of 9 CA1PCs. Gray vertical bars mark the timings of the synchronous ensembles on all traces. Bottom: animals' speed recorded simultaneously with voltage imaging.

(B) LFP ripple power and spiking probability aligned to synchrony ensembles and ripple events. Upper panel: color-coded LFP power at ripple frequency (120-240Hz) aligned to the timings of 64 synchronous ensembles (left) and 12 ripple events (right). Middle panel: Average LFP power of ripple frequency aligned to the timings of the synchronous ensembles (left) and the ripple events (right). Lower panel: Average spike counts of all cells aligned to the timings of the synchronous ensembles (left) and ripple events (right).

(C) Percentages of co-occurrences between population synchrony and ripples. The co-occurrences in the "sync to ripples" group represent the percentages of ripple episodes co-occurring with synchronous ensembles in the same episodes. The co-occurrences in the "ripples to sync" group indicate the percentages of synchronous ensembles co-occurring with ripple episodes. Each gray circle represents an individual session.

(D) Distribution of the ripple modulation indexes ($n=41$ cells). Gray bar: modulation indexes with significant p values ($n=35$ cells); white bars: modulation indexes without significance ($n=6$ cells).

during immobility, the LFP power at ripple frequency (120-240 Hz) exhibited no discernable peaks within any of the synchronous events (**Figure 3B**, left upper and middle panels). Nevertheless, the same LFP power displayed substantial peaks during every ripple event (**Figure 3B**, right upper and middle panels). A striking contrast between synchronous events and ripples was also evident when spiking probabilities during these events were examined. During population synchrony, the mean spiking probability averaged across all cells was high, while the spiking probability was almost zero during ripple events (**Figure 3B** lower panel). In the three sessions where ripples were identified, we calculated the percentages of ripple events coinciding with synchronous ensembles and vice versa. Remarkably, there was no co-occurrence of ripples and synchronous ensembles (**Figure 3C**). Furthermore, we quantified ripple-modulated spiking by calculating the ratio of the difference in mean firing rates during in-ripple and out-ripple periods to the sum of the mean firing rates during both periods. Indexes of most cells were negative, and a striking 85% of cells displayed ripple modulation indexes of -1 , indicating complete suppression of their spiking activities during ripples (**Figure 3D**). Taken together, during novel exploration, synchronous ensembles involved many neurons with millisecond-synchronized spikes and occurred outside the time frames of ripples.

It was puzzling that these CA1PCs exhibited robust spiking activities outside of ripples yet generated few spikes during ripples. To further investigate neuronal activities during ripples, we established a recording condition that allowed us to capture more ripple episodes. Specifically, we immobilized mice in a tube to promote behaviors favoring ripple generation. The mice were habituated to head fixation in a tube in a room distinct from the one where imaging experiments were conducted. On the imaging day, the mice were introduced to the recording room and head-fixed under the microscope for the first time.

CA1PCs were labeled *in utero* on embryonic day (E) 14.5 ($n=56$ cells from 3 sessions in 3 mice) and E17.5 ($n=55$ cells from 3 sessions in 3 mice) and imaged in adult brains. Both neuronal populations exhibited prominent peaks in their grand average CCGs and significantly higher synchronous event rates compared to jittered data (**Figure 3-figure supplement 2A, B**). Approximately 40% of the recorded neurons participated in synchronous ensembles, indicating robust synchronous activity involving a substantial proportion of the recorded cells (**Figure 3-figure supplement 2C**).

In total, 1052 synchronous ensembles and 174 ripple episodes were detected across these imaging sessions. Consistent with findings from walking animals, few synchronous ensembles occurred during ripples when animals were immobilized in a tube (**Figure 3-figure supplement 3A, B**). Moreover, no distinguishable ripple oscillations were observed in synchronous events, and the average firing rates during ripple episodes were near zero (**Figure 3-figure supplement 3C, D**). At the single-cell level, 90% of neurons showed significant negative spiking modulation during ripples, with ripple modulation indexes close to -1 , indicating strong suppression of spiking (**Figure 4Ai**). This suppression extended to subthreshold membrane potentials, as nearly all cells exhibited decreased fluorescence during ripples compared to baseline (**Figure 4Bi, Ci**). These results demonstrate that spiking activity and subthreshold membrane potentials are robustly suppressed during ripples.

Contextual novelty plays a critical role in shaping hippocampal neuronal activities^{37–43}. To assess its influence, we trained mice to become familiar with the imaging procedure and the recording environment over five days and recorded CA1PC activities on the final day. Mean firing rates were significantly reduced in the familiar group compared to the novel group (Familiar group: 1.1 to 5.2 Hz (25th-75th percentiles), median=2.3 Hz, $n=66$ cells, 6 sessions, 4 mice; Novel group: 1.7 to 6.0 Hz (25th-75th percentiles), median=4.2 Hz, $n=111$ cells, 6 sessions, 6 mice, $p=0.0083$, Wilcoxon signed-rank test). Additionally, 15% of the neurons in the familiar group exhibited significantly positive spiking modulation by ripples, while fewer cells showed negative modulation compared to the novel group (**Figure 4A**). During ripples, neurons in the novel group

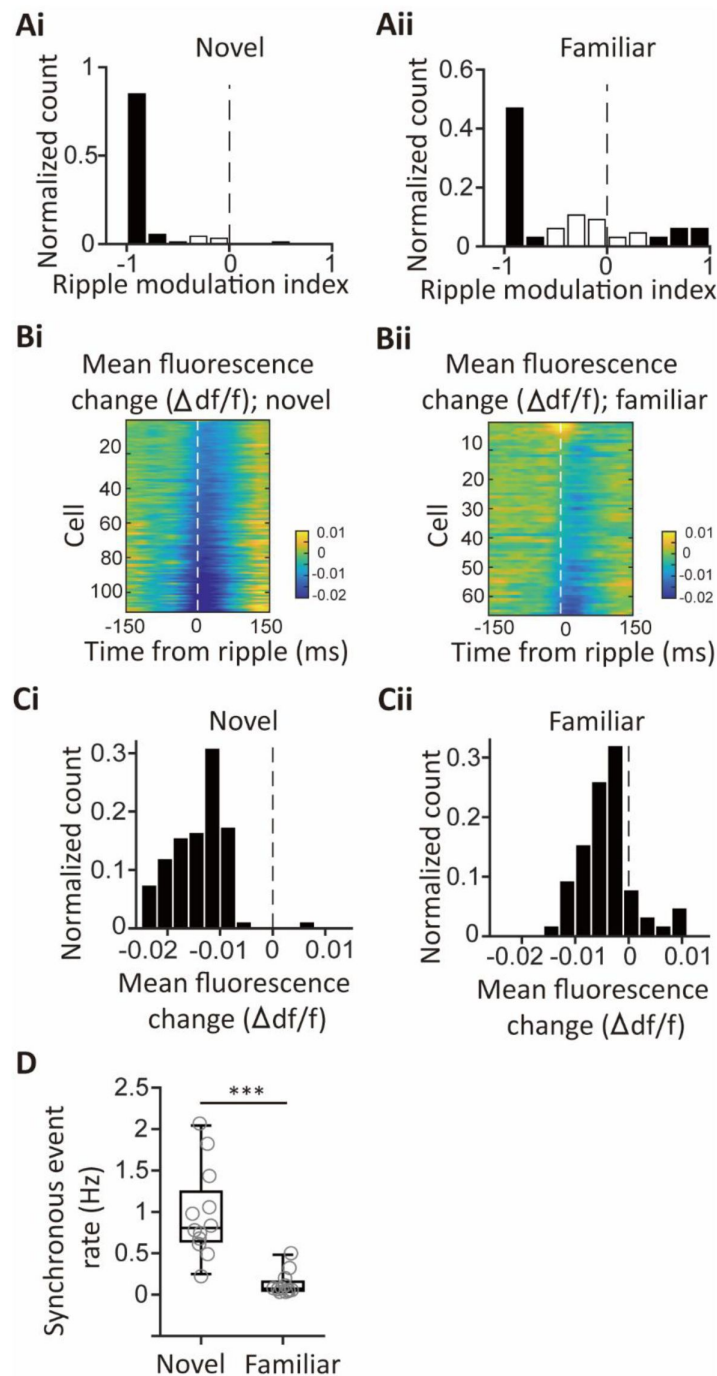


Figure 4.

Comparison of ripple-associated neuronal activities and synchronous event rates between novel and familiar recording contexts.

(A) Histograms of ripple modulation indices for the novel (Ai) and familiar (Aii) recording contexts. Gray bars represent modulation indices with significant p -values, while white bars indicate non-significant modulation indices.

(B) Color-coded representations of mean fluorescence changes relative to baseline fluorescence for individual neurons, averaged across ripple events, in the novel (Bi) and familiar (Bii) recording contexts.

(C) Histograms of mean fluorescence changes within a ± 10 -ms window centered on the peak power of ripple events. Ci corresponds to the novel context, while Cii corresponds to the familiar context.

(D) Boxplots of synchronous event rates for the novel (Di) and familiar (Dii) recording contexts. Mean event rates were 0.98 ± 0.16 Hz for the novel context and 0.13 ± 0.04 Hz for the familiar context ($t(22)=5.20$, $p<0.001$, Student's t -test). *** $p<0.001$

predominantly displayed hyperpolarizing membrane voltage responses, whereas a subset of neurons in the familiar group exhibited prominent depolarizing responses (**Figure 4B**). The mean fluorescence changes in the familiar group shifted toward depolarization compared to the novel group (**Figure 4C**). Finally, synchronous event frequencies were significantly lower in the familiar context, indicating weaker synchronous activities under familiar conditions (**Figure 4D**). These results demonstrate that hippocampal neuronal activities, particularly synchronous ensembles, are strongly influenced by contextual novelty.

Synchronous ensembles are associated with theta oscillations

To compare the network dynamics associated with synchronous ensembles during immobility and locomotion, we aligned LFP traces triggered to the timings of synchronous ensembles. The average LFP traces of a representative session showed prominent theta oscillations around the timings of both immobility and locomotion population synchrony (**Figure 5A**). Spectral analysis of the mean synchrony-triggered LFP traces during immobility and locomotion revealed theta frequency components of 4–12 Hz, with a slightly higher peak frequency in locomotion LFP compared to immobility LFP (**Figure 5B**). Furthermore, we computed LFP theta modulation and identified preferred phases of synchronous ensembles. In line with the previous result, synchronous ensembles displayed a high level of modulation by LFP theta oscillation (modulation strength: 0.61 ± 0.05 during immobility and 0.74 ± 0.04 during locomotion, $n=5$ sessions). Although the preferred phases varied from session to session due to differences in recording sites along the proximal-distal axis of the hippocampus, the timings of individual ensembles were consistently locked to the preferred phase of each session (**Figure 5C**). These results establish a strong link between synchronous ensembles and LFP theta oscillation.

Subthreshold membrane voltage (subVm) is one of the critical parameters supporting the generation of action potentials. To investigate the subthreshold signals underlying population synchrony, we aligned subVm traces with the timings of synchronous ensembles and observed that both immobility and locomotion synchrony rode on a subVm waveform oscillating at theta frequency (**Figure 5D**). To elucidate further the relationship between subVm theta oscillation and spikes involved in population synchrony, we calculated the subVm theta modulation and preferred phases of spikes. Spikes participating in both immobility and locomotion synchrony were strongly coupled to the subVm theta oscillation of neurons, with the mean preferred phase being in the latter half of the rising phase closer to subVm theta peaks (**Figure 5E**). Conversely, spikes not participating in synchronous ensembles had weaker modulation strength compared to spikes involved in synchrony (**Figure 5-figure supplement 1A, B**). Furthermore, the membrane voltages triggered by these two categories of spikes showed a significant difference in the hyperpolarization phase of the theta oscillation. Synchronous spikes were situated within cycles of theta oscillations featuring more hyperpolarized troughs than spikes outside synchronous ensembles (**Figure 5-figure supplement 1C, D**). Thus, compared to spikes not participating in synchronous ensembles, synchronous spikes showed a more robust modulation by theta oscillation of membrane voltage during immobility and locomotion in the context of novel exploration.

Notably, phase locking of spikes to their subVm theta oscillation and the synchronous spikes among many neurons suggest that subVms of different neurons would be correlated at the theta frequency. To explore the possibility, we divided the 180-second subVm traces into 1-second segments with 50% overlap and calculated the cross-correlation and magnitude-squared coherence function for each pair of cell segments. Segments were sorted into the immobility or locomotion group based on the animals' speed during each segment (kept below 1 cm/s for immobility and above 3 cm/s for locomotion for more than 90% of the time). In total, we analyzed 511 cell pairs from 5 sessions. Indeed, the cross-correlation of subVm between cell pairs revealed a high correlation at zero lag and periodic modulation of the correlation at theta frequency (**Figure 5F**). Furthermore, the subVm theta coherences were increased in both groups (immobility: 0.50, locomotion: 0.40, median, $n=511$ pairs) and were negatively correlated with soma distances

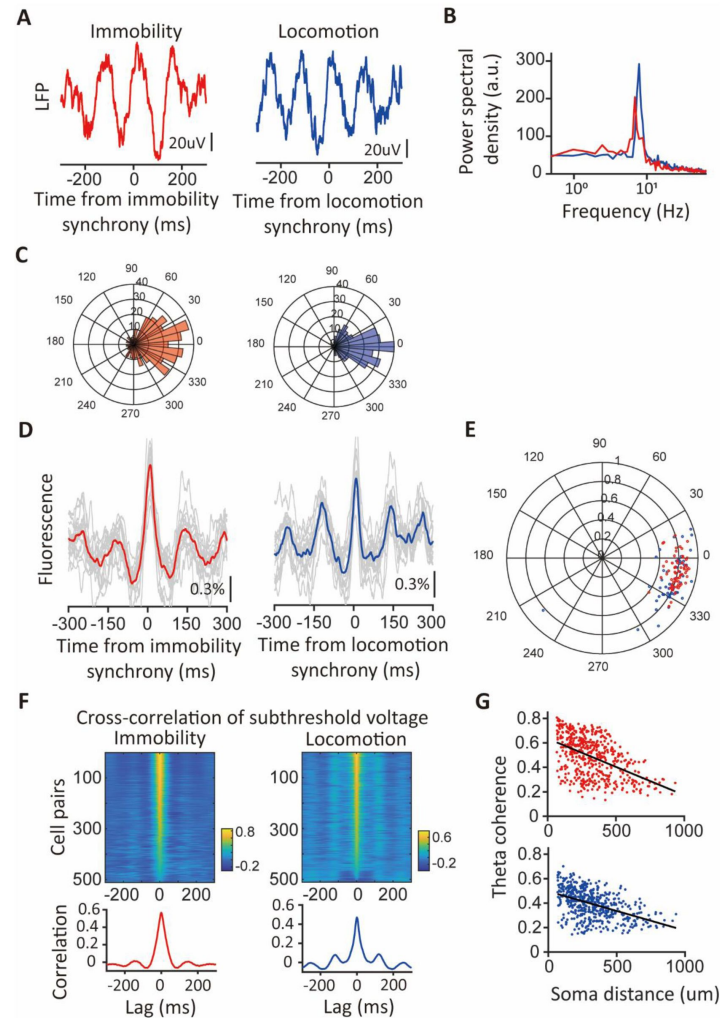


Figure 5.

Theta oscillations in the LFP and subthreshold membrane potential were linked to population synchrony.

- (A) LFP waveforms triggered by timings of synchronous ensembles. Red: triggered by timings of immobility synchrony; blue: triggered by timings of locomotion synchrony.
- (B) Power spectral densities of the triggered LFP waveforms. Red: immobility; blue: locomotion.
- (C) Distributions of the differences between LFP theta phases of individual synchronous ensembles and their preferred phases of the session. Note that most ensembles show minor differences within ± 30 degrees. Red: immobility; blue: locomotion.
- (D) Fluorescence traces triggered by timings of synchronous ensembles. Traces with spikes are excluded. Left panel: traces triggered by timings of immobility synchrony. Right panel: traces triggered by timings of locomotion synchrony. Thin gray lines: mean triggered fluorescence traces averaged across triggers for representative cells. Thick red and blue lines: mean triggered fluorescence traces averaged across cells.
- (E) Polar plot illustrating theta modulation of spikes participating in the immobility synchrony (red dots) and of spikes participating in the locomotion synchrony (blue dots). The angle of the dots indicates the preferred phase, and the distance from the center of the circle represents the modulation strength. Each dot represents the averages from a neuron.
- (F) Pairwise cross-correlation of the subthreshold membrane voltages (subVm). Top: color-coded cross-correlation of subVm between cell pairs during immobility and locomotion periods. Bottom: averaged cross-correlation over all cell pairs for immobility (red) and locomotion (blue) subVm. Both show a central peak at zero lag flanked by theta oscillation.
- (G) Scatter plots of theta coherence against soma distances of immobility subVm (red) and locomotion subVm (blue) between cell pairs.

(immobility: -0.46 , locomotion: -0.48 , $n=511$ pairs, **Figure 5G**). Taken together, during novel exploration, CA1PCs generate synchronous activity associated with theta oscillation and exhibit correlated and theta-coherent subVms.

Correlated intracellular theta and theta-phase locking of the synchronous ensembles raise the question of whether population synchrony among CA1PCs extends beyond synchrony derived from these effects. To address this, we analyzed population synchrony after randomizing the theta cycles during which neurons spiked, while keeping their theta phases unchanged. **Supplementary Figure 4** illustrates a significant reduction in synchronous event rates following theta cycle randomization. The finding indicates spiking at specific theta cycles plays a major role in driving population synchrony.

To further investigate synchronous ensembles across different datasets, we analyzed publicly available hippocampal recordings ‘hc-11’ from the CRCNS repository^{43,44}, where rats navigated novel mazes for water rewards (see Method). Using our algorithm, we identified a significant number of synchronous ensembles during the first three minutes of novel navigation. On average, the rates of synchronous events were 6.4-fold higher than those detected in jittered controls (mean event rate: 2.0 ± 0.3 Hz for the original data vs. 0.32 ± 0.03 Hz for jittered data, $n = 8$ sessions, $p = 0.0078$, $W = 36$, Wilcoxon signed-rank test; **Supplementary Figure 5A**). To assess whether ripple oscillations were associated with these synchronous ensembles, we analyzed ripple event rates and their relationship to population synchrony. During this period, ripple events were infrequent (mean ripple rate: 0.02 ± 0.01 , $n = 8$ sessions), and ripple power peaked during ripple episodes but remained low at the timings of population synchrony (**Supplementary Figure 5B**). Nevertheless, LFP traces aligned to population synchrony revealed prominent theta oscillations (**Supplementary Figure 5C**).

Synchronous ensembles were modulated by LFP theta oscillation (modulation strength: 0.30 ± 0.04 , $n = 8$ sessions, $p < 0.001$), and the timings of individual ensembles were consistently locked to the preferred phase of each session, suggesting a functional coupling of synchronous ensembles to theta oscillations important for information processing (**Supplementary Figure 5D**).

Synchrony between place cells with distinct place tunings

When animals explore a novel maze, spatially tuned spiking activity can form within a few minutes^{45,46}. However, how synchronous ensembles organize their spatially tuned activities during novel exploration remains to be determined. Synchronous ensembles, generating spikes coincidently in short intervals, might display correlated spiking at longer time windows that match behavioral timescales of exploration, potentially leading to shared place responses. On the other hand, CA1PCs are interconnected through inhibitory interneurons^{47–49}. Modulation of activities in inhibitory interneurons by some place cells may influence the activities of other place cells, resulting in divergent spatial tunings^{45,50,51}.

To compare the spatial tunings of synchronous cell pairs, we first calculated the spatial tuning curves of individual neurons by computing the mean firing rates in every spatial bin. On average, 40% of cells displayed selective spatial tunings of their spiking activities and were qualified as place cells⁵² (**Figure 6A**). Among place cell pairs, some showed similar spatial tunings, while others did not (**Figure 6B**). For instance, in an illustrative session depicted in **Figure 6B**, the seventh and eighth place cells had highly similar spatial tunings with a correlation coefficient as high as 0.7. In contrast, the fifth and tenth place cells exhibited vastly different spatial tunings with a correlation coefficient as low as -0.5 .

We further explored the relationship between the similarity of spatial tunings and synchronization strength among pairs of place cells. We measured synchronization strength as the peak of the CCG normalized by the averaged jittered spike count within the same bin as the CCG

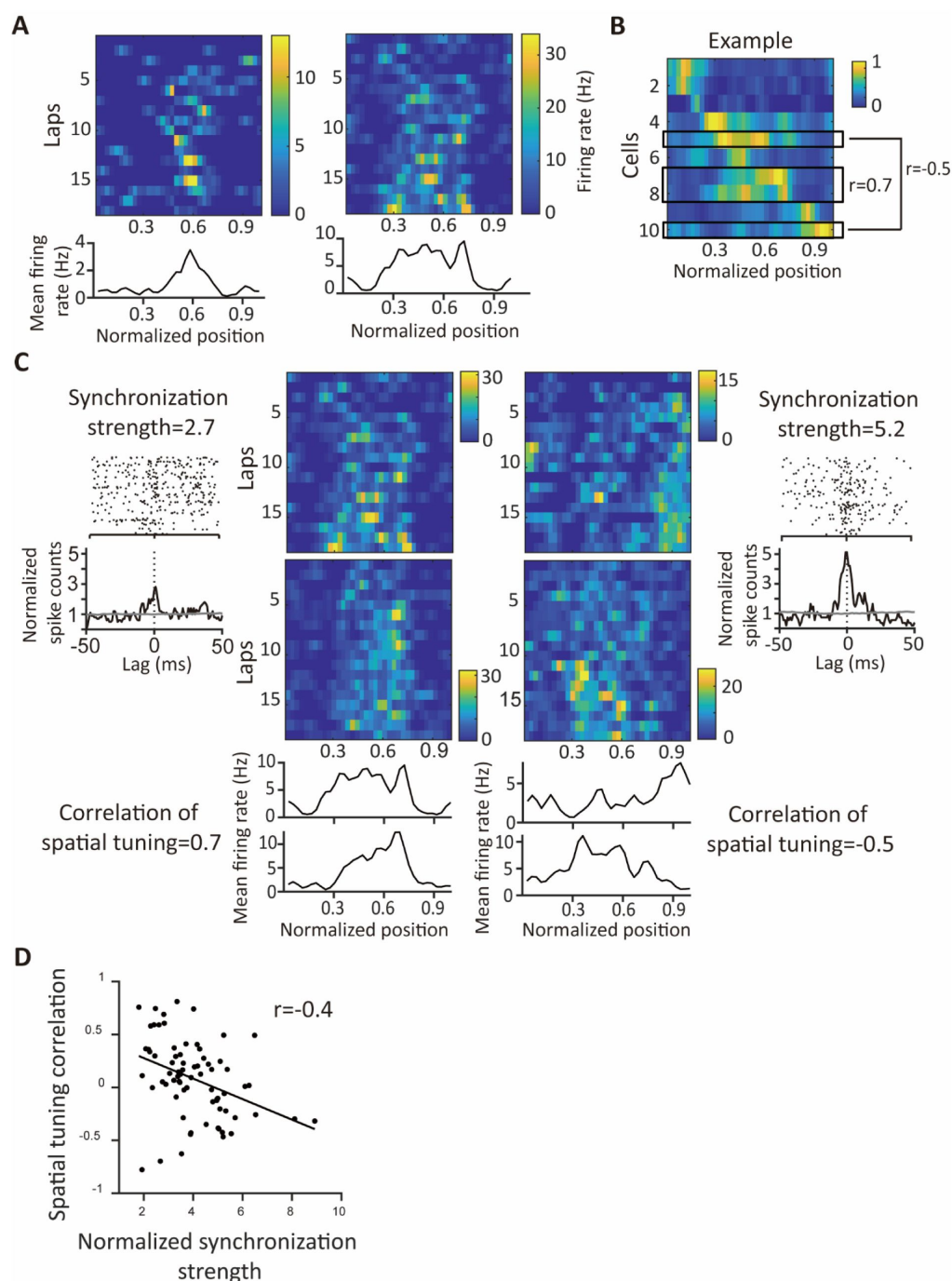


Figure 6.

Negative correlation between spatial tuning similarity and synchronization strength during novel exploration.

(A) Spatially binned rate maps of place cells. High firing rates are indicated in yellow, and low firing rates are noted in blue. Bottom: the mean firing rates along spatial bins averaged across laps.

(B) Spatial tunings of 10 place cells from an example session. Cells are sorted based on the locations of their peak firing rates.

(C) Two example pairs of neurons with cross correlograms during locomotion and spatially binned firing rate maps.

(D) A scatter plot of correlation coefficients of spatial tunings against normalized synchronization strengths for place cell pairs. Each dot represents a pair of place cells. Synchronization strengths are negatively correlated with similarities of spatial tunings (Spearman correlation coefficient=-0.4, $p<0.001$, $n=73$ pairs).

peak. We segregated spikes based on these behavioral states to accommodate differences between locomotion and immobility and calculated the corresponding CCG and synchronization strength.

During locomotion, we observed a robust negative correlation between synchronization strength and the similarity of spatial tunings among cell pairs. In other words, cell pairs with more precisely co-active spikes during locomotion exhibited more distinct place-tuning profiles (**Figure 6C, D**). This anti-correlation between synchronization strength and place tunings indicates coordination between place cells' temporal and spatial coding. On the other hand, synchronization strength during immobility showed little correlation with the similarity of spatial tunings among place cell pairs, suggesting that correlated activities at the millisecond timescale do not extend to slow, behavioral-relevant timescales (Figure 6-figure supplement 1). Thus, place cells with distinct place fields are linked by synchronous activity during novel locomotion.

Discussion

Our findings reveal the presence of synchronous ensembles comprising a substantial number of CA1PCs during both locomotion and immobility in a novel environment. These ensembles concurrently engage neurons representing diverse aspects of the environment and are rarely observed during ripple events. Instead, they closely align with LFP and intracellular theta oscillations. Our data demonstrate theta-associated synchronous ensembles during spatial memory acquisition, coordinating numerous CA1 place cells that transmit information about different segments of the environment.

The correlation between observed population synchrony and theta oscillations is intriguing. Theta oscillations are typically associated with the sequential activation of neurons over time, resulting in reduced neuronal synchrony²². However, theta oscillations come in different types, each characterized by distinct pharmacological properties and behavioral correlates^{53–55}. For example, type 1 theta, or translational theta, occurs during voluntary movements and is sensitive to N-methyl-D-aspartate receptor (NMDAR) blockers. On the other hand, type 2 theta, or attentional theta, is slightly slower and is blocked by muscarinic receptor antagonists, emerging during states of arousal or attention, such as when entering a new environment^{56–58}. Consistent with these distinctions, the peak of the power spectrum density shows a distinctively slower theta frequency during immobility compared to locomotion (**Figure 5B**).

During theta oscillations, cholinergic neurons in the medial septum excite downstream CA1PCs, leading to increased firing rates and membrane potentials^{59–61}. Given the widespread influence of cholinergic inputs on numerous CA1PCs, theta-coupled cholinergic inputs may drive depolarization and firing across many CA1PCs. Additionally, they could enhance the likelihood of synchronicity. Furthermore, GABAergic neurons in the medial septum, activated during theta oscillations, could activate interneurons in the CA1 region^{55,62–63}. These CA1 interneurons, in turn, have the capability to entrain the activities of pyramidal neurons at both sub- and suprathreshold levels, thereby promoting synchronization among pyramidal neurons^{64,65}. In summary, cholinergic and GABAergic neuron activation during theta waves may contribute to the excitation of CA1PCs and the consequent increase in their synchronicity.

Theta oscillations have long been implicated in memory acquisition. Specifically, theta power correlates positively with learning performance^{66–68}. Perturbing generation of theta waves leads to impairment in memory acquisition^{69–73}, while electrical stimulation restoring theta rhythms rescues learning performance^{74,75}. Furthermore, increased spike-theta coherence correlates with better memory, and stimulations that boost theta phase-locking of spikes enhance memory^{76–78}. Despite abundant evidence supporting a link between theta oscillations and memory, the precise network mechanism by which theta oscillations support memory acquisition remains unclear. Theta oscillations are hypothesized to link spatially distributed neurons into

functional ensembles to support memory acquisition.¹³ The identified synchronous ensembles support the hypothesis, providing a network substrate that links theta rhythms to the initial stage of memory acquisition.

Recent studies have demonstrated elevated coactivity of hippocampal principal cells during theta-associated dentate spikes, events that are temporally coupled to theta oscillations and distinct from sharp-wave ripples.^{79–81} These findings highlight a network state characterized by rich synchronous ensembles associated with theta, but not ripple, oscillations—a pattern consistent with our results. Our analyses of publicly available data further support these observations, revealing theta-associated population synchrony in a different dataset (**Supplementary Figure 5**). These results suggest that hippocampal neurons achieve synchrony through network mechanisms distinct from ripple-related events, reinforcing the hypothesis that this phenomenon is generalizable across experimental conditions and behavioral contexts.

Previous studies have reported that a small percentage of CA1PCs generate spikes during ripple episodes.^{8,14,17} These measurements are typically conducted in familiar environments. In contrast, when animals are introduced to a novel environment, ripple-associated spiking activities are strongly suppressed, and CA1PCs exhibit prominent hyperpolarizing responses in their membrane potentials (**Figure 3**, **4Ai–Ci**, and **Figure 3-figure supplement 3**). Interestingly, after familiarization training, a larger proportion of CA1PCs display positive ripple modulation of spiking activities and depolarizing responses associated with ripples (**Figure 4Aii–Cii**). Consistent with the observation of ripple-associated spiking activity in familiar environments, our study underscores the influence of contextual novelty on hippocampal dynamics. Specifically, brain states associated with novelty may modulate the activity of neurons upstream of CA1PCs, driving their membrane potentials toward hyperpolarization during ripples. While the reduced correlation between ripples and spikes in novel environments may initially appear counterintuitive, this unique neuronal activity pattern likely reflects adaptive changes in hippocampal circuits related to the animals' behavioral states, such as heightened vigilance in novel environments.

Although CA1PCs are known to exhibit weaker recurrent connectivity than that of CA3 pyramidal neurons, CA1PCs did communicate through monosynaptic connections.⁸² These connections can be dynamically adjusted by the spike-timing-dependent plasticity mechanism^{83–85}, and therefore could be affected by the transient population synchrony. In addition, CA1PCs are known to communicate through inhibitory interneurons. Multiple pyramidal neurons converge onto a single interneuron, and individual connections are often weak and unreliable.^{48,86,87} This connectivity scheme requires multiple pyramidal neurons spiking synchronously within the time window of synaptic integration to effectively activate postsynaptic neurons.⁴⁸ Furthermore, population synchrony among CA1PCs may propagate to downstream areas, synchronizing neurons outside the hippocampus. With their large ensemble sizes, these synchronous ensembles may synergistically convey relevant information or reorganize the connectivity of the neuronal networks.

Prior studies in sensory systems have commonly revealed that neurons with synchronous activities typically share similar sensory responses.^{88,89} However, the synchronous ensembles we have identified often encompass neurons with distinct preferences for place fields. Pairwise synchrony demonstrates a negative correlation with place field overlap (**Figure 6**). These findings indicate that synchrony does not necessarily imply similar tunings between neurons. Furthermore, these synchronous ensembles involve place cells encoding different spatial information. Recognizing the crucial role of neuronal synchrony in binding distributed neurons in the brain^{3,5}, population synchrony may serve as a mechanism to integrate diverse features of the same environment into a cohesive mental map.

Methods

Animal

Time-pregnant wild-type C57BL/6J female mice underwent in utero virus injection of CamkII-cre virus (105558-AAV1, Addgene) into the left side of the lateral ventricles. The resulting offspring were raised in a breeding room under controlled temperature and humidity conditions and a 12-hour light/dark cycle (lights on from 7:00 to 19:00). The mice used in the experiments were between 12 and 18 weeks old at the time of the recordings. All surgical procedures, behavioral training, and recording protocols were approved by the NYCU Institutional Animal Care and Use Committee.

In-utero virus injection

Time-pregnant mice on embryonic day 14.5 or 17.5 were anesthetized with isoflurane (induction:4-5%; maintenance:1-3%) before surgery. The animals were placed on a heating pad in a supine position to maintain body temperature. Aseptic procedures were implemented to maintain sterile conditions during the surgery. Uterine horns were exposed one at a time using spoons and fingers. Warm and sterile phosphate-buffered saline (PBS) was used to rinse the uterus to prevent it from drying out. Micropipettes loaded with CamkII-cre virus solution were used to inject 0.2 μ L of virus suspension into the left side of the lateral ventricle of each embryo. The injection resulted in Cre expression among neurons born on the day of injection, with earlier injection labeling neurons located on the deeper side of the cell layer. The uterine horn was then gently returned to the abdominal cavity. The exact process was repeated to the other uterine horn. After the surgery, ketorolac (6mg/kg) and cefazolin (1g/kg) were administered for two days to minimize pain and inflammation.

Cranial window and cannula implantation

The offspring of the mice that had undergone in-utero injection were subjected to chronic hippocampal window surgery when they were between 2 and 4 months old. The animals were initially anesthetized using isoflurane (induction:4-5%; maintenance:1-3%). Topical anesthesia was applied by administering 0.5% lidocaine to the wound margins. After achieving deep anesthesia, an incision was made in the skin, and a circular section of the skull was removed, centered 2 mm caudal and 1.8 mm left to the bregma. Using 30-gauge needles and forceps, the dura was removed from the exposed area, and cortical tissue within the craniotomy was aspirated. A serial dilution experiment was conducted to determine an optimal titer of the virus carrying Voltron2 genes, minimizing cell toxicity, for use in this and in previous imaging experiments²⁹. A fine injection pipette (tip diameter 10-60 μ m) was used to inject AAV2/1-CAG-flex-Voltron2-ST (2.7×10^{12} GC/ml, a generous gift from Dr. Eric Schreiter and the GENIE team at HHMI Janelia Research Campus) into the exposed regions at a depth of 200 μ m (up to six injection sites and 100-200 nL of viral suspension). An optical chamber was constructed by placing a cannula with a cover slip attached at the bottom over the craniotomy and sealing it with dental acrylic or C&B Metabond (Sun Medical). A custom-built titanium frame was then cemented to the animal's head using dental acrylic or C&B Metabond. Mice received ketorolac (6mg/kg) and cefazolin (1g/kg) for two days following surgery to minimize pain and inflammation.

Behavioral training

After at least two weeks of recovery from the window surgery, the mice underwent behavioral training to ensure they were calm and attentive in the test environment. For mice subjected to imaging experiments where they explored an air-lifted plastic track, the initial training phase involved acclimating the mice to head fixation and treadmill running, which took place over 3-5

days in a separate room distinct from the recording setup. Following this training, the imaging experiments were conducted. On the day designated for imaging, the mice were introduced to the recording room for the first time just before the imaging session. During the imaging experiments, the mice were securely held in a head-fixed position beneath the microscope and on an air-lifted plastic track that rested on an air table (AirLift, Neurotar; **Figure 1A**). The track was adorned with various shapes of signs to help the mice navigate the track. Subdued blue lighting was provided to illuminate the track. Before the initiation of imaging, the mice were confined in a corner of the track using gates. Subsequently, the gates were opened upon the start of imaging, allowing the mice to explore the new environment during their initial laps (**Figure 1A** Top view). The Mobile HomeCage tracking system (Neurotar) was used to track the mice's position and speed. During the acquisition of image frames, TTL pulses were sent by the high-speed camera to the Mobile HomeCage tracking system to facilitate data alignment. Mice subjected to imaging experiments with head fixation in a tube were first trained in a separate room, distinct from the recording setup, for 1-2 days to habituate them to the head fixation procedure. Following this training, the mice were introduced to the recording room for the first time immediately before the imaging session. During imaging, the mice were securely positioned in a head-fixation setup within an acrylic tube and remained stationary. To further familiarize the mice with the recording context, the imaging procedure was repeated for 4 days without injecting the JF-HaloTag ligand. On the fifth day, the JF-HaloTag ligand was injected prior to imaging, and voltage imaging was performed following the same procedure as in the previous sessions.

Voltage imaging

JF₅₅₂-HaloTag ligand⁹⁰ (a generous gift from Dr. Luke Lavis) was first dissolved in DMSO (20 μ l, Sigma) and then diluted in PluronicTM F-127 (20 μ l, P3000MP, Invitrogen) and PBS to achieve a final concentration of 0.83 mM of JF₅₅₂-HaloTag ligand. The solution was then injected intravenously through the retro-orbital sinus. Imaging sessions were initiated 3 hours after the injection of the JF₅₅₂-HaloTag ligand. Fluorescence sensors were excited using a 532-nm laser (Opus 532, Laser Quantum) with an excitation filter (FF02-520-28, Semrock), and the emitted light was separated from the excitation light using a dichroic mirror (540lpxr Chroma) and passed through an emission filter (FF01-596183 Semrock). A 16X, 0.8 NA objective (Nikon) was used to collect the emission light which was then imaged on a CMOS camera (DaVinci-1K, RedShirt) with a 50-mm camera lens (Nikkor 50mm f1.2, Nikon) as the tube lens. Images were acquired using Turbo-SM64 (Sci-Measure) at 2kHz with a resolution of 190 x 160 pixels, corresponding to a field of view of 1.4 x 1.2 mm. The number of frames in an image session was set at 360,000, corresponding to a duration of 180 seconds per session. Imaging depths ranged between 70 and 170 μ m from the window surface, where the CA1 pyramidal neurons were located. Time-lapse images were collected, and image stacks of the same field of view along the z-axis were acquired. The image stacks consisted of 100 plans separated by two μ m, covering the depth from the window surface to 200 μ m from the surface.

LFP recording

The LFP signal from the contralateral CA1 region of the hippocampus was simultaneously recorded during voltage imaging. To implant the LFP electrode, a small craniotomy was performed at the dorsal CA1 coordinate (2mm caudal and 1.8 mm right to the bregma). A tungsten electrode (#100211, 38 μ m in diameter, polyimide insulated, California Fine Wire) was inserted into the dorsal CA1 region, and the wire and pins were cemented in place with dental acrylic. The electrical signal was amplified and filtered (1-1kHz) using a DAM80 amplifier (World Precision Instruments). The amplified signal was acquired by an I/O device (National Instruments USB-6341) and recorded using WaveSurfer (Adam Taylor, HHMI Janelia Research Campus) at 10 kHz during implantation and TurboSM64 (RedShirt) at 8 kHz during imaging.

Imaging somata and dendrites *in vivo* using a two-photon microscope

Immediately after voltage imaging, Voltron-labeled neurons were examined *in vivo* under a two-photon microscope to assess the integrity of their membranes and dendrites. Images were acquired using a two-photon scanning microscope equipped with a femtosecond-pulsed laser (Chameleon, Coherent) and a resonant scanner controlled by ScanImage 4.2 software⁹¹ (<http://scanimage.org>). A 16x water-immersion objective (Nikon, 0.8 NA, 3 mm working distance) was used for imaging. Z-stacks of images (512 x 512 pixels, 600μm x 600μm) were collected with excitation light at 800 nm. The emission light was separated from the excitation light using a primary dichroic mirror (FF735-Di02-25×36, Semrock), followed by a secondary dichroic mirror (565dclp, Chroma) and an emission filter (FF01-609/57-30-D, Semrock). Each image stack consisted of 100 planes, separated by 2μm, covering depths from the window surface to 200 μm below the surface.

Histological processing and quantification of soma depth

Mice with Voltron₂₅₅₂ labeling were anesthetized with intraperitoneal injections of ketamine (250 mg/kg) and xylazine (25 mg/kg) and transcardially perfused with 4% paraformaldehyde in phosphate-buffered saline (PBS; 2 mL/g). The brains were post-fixed in 4% paraformaldehyde overnight and then washed in PBS. Coronal brain sections, 50μm thick, were prepared using a VT1200 vibratome (Leica). Sections were mounted with Vectashield mounting medium containing DAPI (VectorLabs) to visualize nuclei. The soma depth of Voltron-labeled cells was measured relative to the border between the *stratum pyramidale* (SP) and the *stratum radiatum*. Depth values were normalized to the thickness of the SP, with larger depth ratios indicating somata located deeper along the radial axis.

Voltage imaging preprocessing and spike detection

Fluorescence intensities were corrected for brain movement using rigid registration. Regions of interest (ROIs) were manually selected by grouping pixels that cover individual somata. Previous studies have described and validated the procedure for imaging preprocessing and spike detection^{29,92,93}. In short, the fluorescence intensities of individual neurons were calculated by averaging the fluorescence intensities of pixels from the same ROIs. Bleaching was corrected by calculating the baseline fluorescence (F_0) at each time point as an average of the fluorescence intensities within ± 0.5 seconds around the time point. The dF/F was calculated as the F_0 minus the fluorescence intensity of the same time point divided by F_0 . Positive fluorescence transients were detected to identify spikes from the high-passed dF/F traces created by subtracting the dF/F traces from the median-filtered version with a 5-ms window. To simulate the noise of recordings, high-passed dF/F traces were inverted, and the amplitudes of the transients detected from the inverted traces were used to construct a noise distribution of the spike amplitudes. A threshold was set by comparing the amplitudes of the detected transients with the noise distribution of the spike amplitudes to minimize the sum of type I and type II errors. Spikes were first detected when transients were larger than the threshold. Then, spike amplitudes smaller than half of the top 5% spike amplitudes were excluded. The signal-to-noise ratio (SNR) was calculated for each neuron as a ratio of the averaged spike amplitudes over the standard deviation of the high-passed dF/F traces, excluding points 2 ms before and 4 ms after each detected spike to estimate the quality of the recordings. The crosstalk between ROIs was routinely checked by examining the refractory periods of neurons in auto-correlograms. Only neurons meeting the following criteria were included: (1) SNR larger than 5, (2) full width at half maximum of the spike waveform longer than 0.8 ms, (3) mean spike rates higher than 0.1 Hz, (4) distances between soma pairs at least 70 μm. Sessions with at least nine neurons meeting the selection criteria were used for further analysis.

Behavioral states of locomotion and immobility

The periods of locomotion were defined as instances when the animals' speed exceeded 3 cm/s, while periods of immobility were determined as instances when the animals' speed fell below 1 cm/s.

Mean firing rates during locomotion and immobility

To calculate the mean firing rates during locomotion, we counted the number of spikes during locomotion periods and divided this number by the total duration of locomotion. Similarly, for mean firing rates during immobility, we calculated the number of spikes that occurred during immobility periods and divided this number by the total duration of immobility.

Correlation between instantaneous firing rate and animal speed

To estimate the instantaneous firing rates of individual neurons over time, we convolved their spike trains with a Gaussian window of 250 ms. Subsequently, we calculated the Pearson correlation coefficient between these instantaneous firing rates and the animal's speed for each neuron.

Grand average cross-correlogram (CCG)

Grand average CCGs were generated by constructing histograms of relative spike timings between spikes of a reference cell and spikes of all other neurons in the same session as the target cell population. Histograms were binned using 1-ms time bins. Spike counts were normalized by the number of reference spikes times the number of all other neurons. An equal number of cells from different sessions were randomly selected to form the shuffled target cell population to assess the statistical significance of the CCG peaks. Subsequently, grand average CCGs were computed using shuffled spike trains for comparison. Grand average CCGs were calculated using all spikes in the 180-s recording periods, subsets of spikes during immobility periods (immobility grand average CCGs), and subsets of spikes during locomotion periods (locomotion grand average CCGs).

Detection of the population synchrony

To assess population synchrony, we counted the number of spikes from all neurons within a sliding window of 25 ms. To test the significance of the synchrony, we generated surrogate data by jittering spike timings within a ± 75 -ms window 500 times. The means and standard deviations of each sample point were calculated over the jitter population. Population synchrony was detected as peaks when the spike count in the original data exceeded the mean plus four standard deviations of the spike counts in the jittered data. Synchrony was also evaluated in modified spike trains, including those where later spikes in bursts were removed and those where spike times were shuffled across theta cycles while preserving phases. To estimate synchronous event rates, we divided the entire 180-second recordings into halves and calculated the event rate in each of the 90-s segments. Additionally, ensemble sizes were quantified by counting the percentage of neurons participating in each synchronous event.

To detect population synchrony in a publicly available dataset, we analyzed the “hc-11” dataset from CRCNS, which contains hippocampal recordings from rats running on novel mazes for water rewards at designated locations^{43,44}. Spikes recorded during the first three minutes of the novel-maze running epochs were used to identify synchronous ensembles and generate jittered control data. Spike counts for both the original and jittered data were calculated using a sliding 25-ms window. The algorithm for testing the significance of synchrony was the same as described above. Synchronous event rates were estimated using the entire 3-minute spike trains for both the original and jittered data.

Pairwise cross-correlogram (CCG) of neuronal pairs

To compute the CCG, we first determined the time difference between spikes in the target and reference neurons and then binned these values into 1-ms time bins. CCGs were only computed for pairs with more than 100 counts within the CCG window of ± 500 ms to ensure reliable correlation. We defined the peak of the CCG as the local maxima within ± 30 ms of lag. To assess the significance of cross-correlations, we generated surrogate data by jittering the spike times of the target neuron within a ± 75 ms window. We computed the CCG peak for a thousand iterations and calculated p-values by comparing the percentile of the peak in the null distribution of the jittered peak values. The synchronization strength was estimated as the ratio between the original CCG peak and the mean spike count of the same bin as the original CCG peak averaged across the jittered histograms. For display purposes, the CCGs were smoothed using a 5-point moving average. CCGs were calculated using all spikes in the 180-s recording periods, subsets of spikes during immobility periods (immobility pairwise CCGs), and subsets of spikes during locomotion periods (locomotion pairwise CCGs).

LFP ripple analysis

To detect ripples, we performed band-pass filtering of the LFP signal between 120 and 240 Hz, followed by computing the envelope using the absolute values of the Hilbert transform, which was then low-pass filtered at 20 Hz. The start and end times of ripples were identified by setting the upper and lower thresholds at the mean plus 7 and 3.5 times the standard deviation of the envelope, respectively. For a ripple event to be included, three conditions had to be met: (1) values within the event were higher than the lower threshold, (2) at least one value exceeded the upper threshold, and (3) the minimum duration of the event was 30 ms. The start and end times of a ripple event were defined as the positive and negative crossings of the lower threshold, respectively. We defined in-ripple periods as the time between the start and end of detected LFP ripple events, while out-ripple periods were periods outside the in-ripple periods. The envelope peak values of the detected ripple events were used as the time of ripple occurrence. These timings were utilized as trigger timings to calculate averages triggered by ripples. Two co-occurrences were quantified: the percentages of LFP ripple oscillations that occurred with synchronous ensembles and the percentages of synchronous ensembles that occurred with LFP ripple oscillations. The same ripple detection algorithm was applied to the hc-11 dataset. To account for artifacts in the recordings, raw traces and power spectra of the detected events were plotted and inspected for false positives.

Ripple modulation index

We utilized the ripple modulation index to assess changes in firing rates during ripples compared to periods outside ripples. This index was computed as the ratio of the difference in mean firing rates during in- and out-ripple periods to the sum of the mean firing rates during both periods.

Specifically, we first calculated the mean firing rate during in-ripple periods and the mean firing rate during out-ripple periods. Next, we subtracted the mean firing rate during out-ripple periods from the mean firing rate during in-ripple periods and divided the result by the sum of the mean firing rates during both periods. The resulting ripple modulation index indicates the magnitude and direction of the firing rate modulation during ripple compared to periods outside ripples. To assess the significance of the modulation index, we shuffled the spike timings and repeated the calculation a thousand times to generate a distribution of modulation indices from the shuffled data. The original modulation index was then compared to this distribution, and its significance was determined if it exceeded the 95% confidence level of the shuffled indices.

LFP traces triggered by population synchrony and analysis

The midpoints of the synchronous event intervals were utilized as triggers to align LFP traces from both our recordings and the hc-11 dataset. These traces were aligned within a ± 1 -second window around the trigger time points. To calculate the spectral density functions of the aligned LFP traces, we performed a 16,384-point fast Fourier transform of the 2-second LFP traces. The power spectra were then estimated as the squared absolute values of the Fourier coefficients. Ripple power was calculated as the sum of the power distributed between 120 and 240 Hz. To calculate theta modulation strength and preferred angles of the synchronous ensembles, LFP was first filtered within the theta frequency range. Then, the phase $\phi(t)$ and the instantaneous amplitude $A(t)$ of the filtered LFP were computed using the Hilbert transform. A vector $V_k = A(t_k)e^{i\phi(t_k)}$ was assigned to each synchronous ensemble that occurred at time t_k . The modulation strength and the preferred phase angle were determined as the absolute value and the phase angle of the summed vector over all ensembles, respectively. Modulation strengths were normalized by the total length of all vectors.

Subthreshold membrane voltage (subVm)

To compute the subVm of individual neurons, we began by determining the difference between the raw dF/F signal and the high-pass version of dF/F. Next, we calculated the spike threshold for each cell by identifying the voltage value corresponding to the peak of the first derivative of the spike waveforms. Any fluorescence values in the slow dF/F time courses that surpassed the spike thresholds were linearly interpolated.

SubVm traces triggered by population synchrony

The midpoint of the time intervals, during which synchronous ensembles were detected, served as the reference points for aligning fluorescence traces within a ± 300 ms window around those reference points. To eliminate any interference from membrane potential fluctuations associated with spikes, we excluded aligned traces in cases where any spike occurred within the time intervals of the aligned traces.

SubVm theta modulation and phase preference of spikes occurred inside and outside the population synchrony events

The subVm was initially filtered within the theta frequency range. Subsequently, the phase $\phi(t)$ and the instantaneous amplitude $A(t)$ of the filtered subVm were computed using the Hilbert transform. Spikes were categorized into four groups: those occurring during or outside immobility synchronous events and those occurring during or outside locomotion synchronous events. Within each category, a vector $V_k = A(t_k)e^{i\phi(t_k)}$ was assigned to each spike that occurred at time t_k .

The modulation strength and preferred phase angle were determined as the absolute value and phase angle, respectively, of the summed vector over all spikes within a category. To facilitate comparison between neurons, modulation strengths were normalized by the total length of all vectors. The significance of the modulation strength was tested by shuffling the spike timings and recalculating the modulation strength a thousand times to generate a distribution based on the shuffled spike timings. The original modulation strength was then compared to the distribution, with significance determined if it exceeded the 95% confidence interval of the shuffled values. Significant modulation strengths were plotted and compared across groups.

Spike-triggered fluorescence traces

We extracted segments of fluorescence traces using a ± 300 ms time window centered on the spike timings. To examine variations in fluorescence waveforms triggered by spikes within and outside synchronous events, we categorized the fluorescence traces based on whether the spikes occurred

within or outside these events. Subsequently, we performed pairwise comparisons of the fluorescence values from the same neuron, concentrating on spikes occurring during corresponding behavioral states. Neurons with four or fewer triggering events in any of these categories were omitted from the analysis.

Cross-correlation and coherence analysis to the subVm traces

To calculate cross-correlation and theta coherence of subVm between pairs of neurons during locomotion and immobility, we initially divided the 180-second subVm traces into 1-second segments with a 50% overlap. These segments were then categorized into the locomotion group if the animal's speed remained above three cm/s for more than 90% of the time within that segment. Similarly, segments were sorted into the immobility group if the animal's speed remained below one cm/s for more than 90% of the time within that segment. Subsequently, we computed the cross-correlation and magnitude-squared coherence function of frequency for pairs of co-occurred segments belonging to a specific cell pair. Mean cross-correlations of individual cell pairs were averaged across segments that occurred in the same behavioral states (i.e., locomotion or immobility states). To estimate theta coherence, we averaged the values of the coherence function within the theta frequency range (4-12 Hz). Theta coherences from the same cell pair during the same behavioral state were pooled together, and the average was used as an estimate for the theta coherence of that cell pair during a particular behavioral state. Finally, Spearman correlation coefficients were calculated between the mean theta coherences of cell pairs and their soma distances.

Spatial analysis

We partitioned a 90-cm-long track into 36 spatial bins to analyze spatial tuning properties. For each cell, the spatial tuning curve was computed by determining the mean firing rate, defined as the number of spikes within the bin when the animal's speed exceeded three cm/s, divided by the total dwell times using the same speed threshold. The peak firing rates were identified as the maximal rate on the spatial tuning curve. Spatial selectivity was calculated as one minus the circular variance of the spatial tuning curve. Neurons were classified as place cells when their spatial selectivity exceeded 0.25, and peak firing rates exceeded 1 Hz. To assess the similarity of spatial tuning between pairs of neurons, Spearman correlation coefficients were calculated for the spatial tuning curves. To explore the relationship between levels of synchronization and spatial tuning similarity among pairs of place cells, Spearman correlation coefficients were computed between synchronization strengths of cell pairs and correlation coefficients of their spatial tuning curves.

Data analysis and statistics

Data analysis was performed using MATLAB (Mathworks, R2019b), and results were presented as standard error of the mean unless otherwise specified. Boxplots were used to depict the median, first quartile, and third quartile, with the middle line, the upper edge, and the lower edge of the box, respectively. The whiskers of the boxplots represented the 5th and 95th percentiles of the distribution.

Supplementary figures

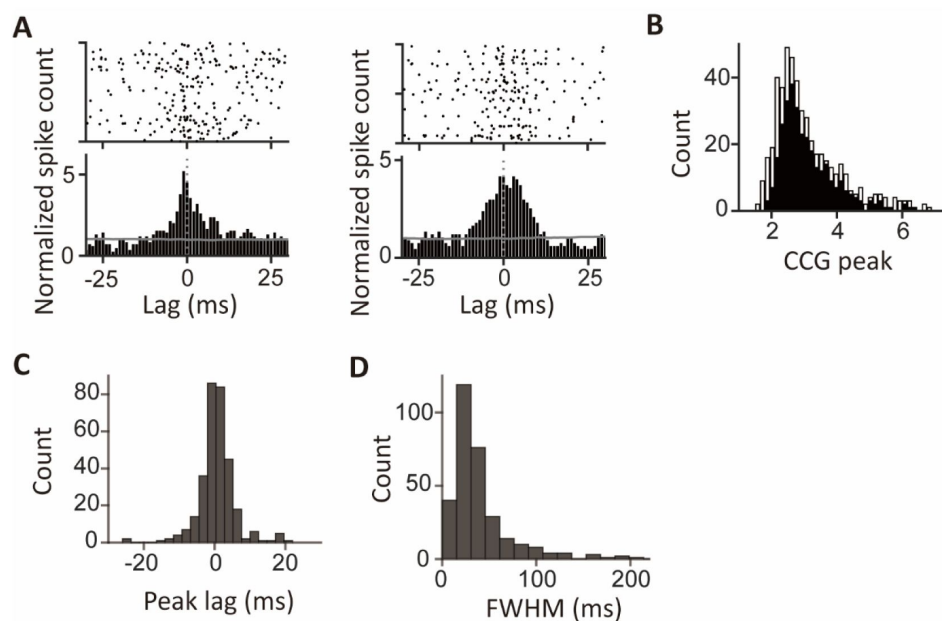


Figure 1-figure supplement 1.

Pairwise cross-correlogram (CCG) between CA1 pyramidal cells.

(A) CCGs of two representative cell pairs exhibiting significant peaks compared to the jittered spike trains of the target cells (gray lines).

(B) Distribution of the CCG peaks (n=497 pairs). Black: CCG peaks with significance ($p < 0.05$). White: CCG peaks without significance ($p \geq 0.05$).

(C) Histogram of peak lags of significant CCG peaks (n=315 pairs).

(D) Histogram of FWHMs of significant CCG peaks (n=315 pairs).

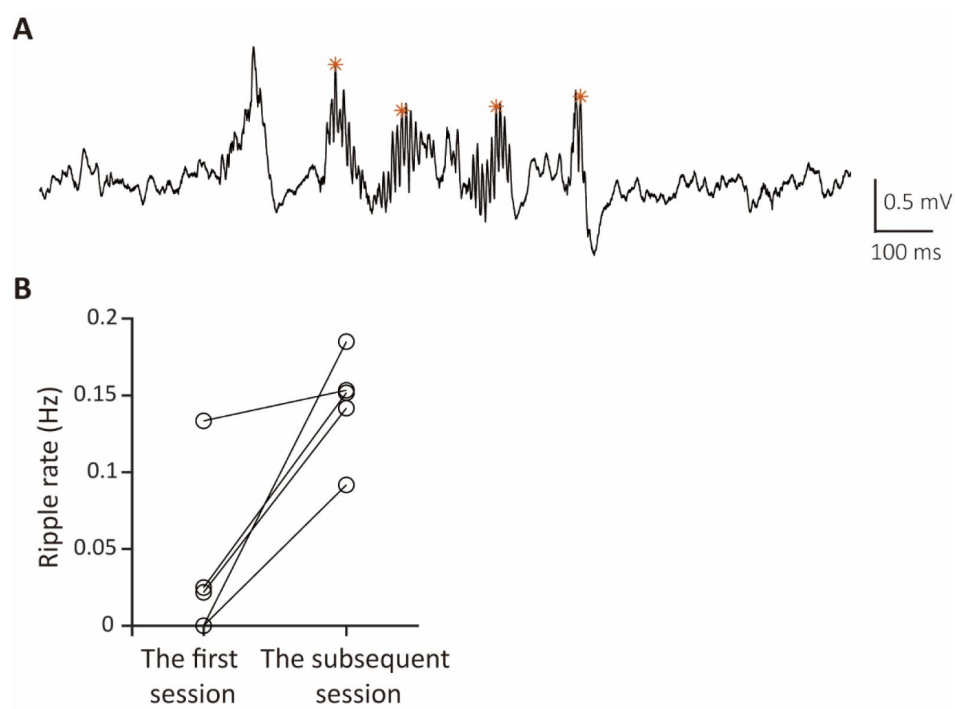


Figure 3-figure supplement 1.

LFP ripples recorded in the first and subsequent sessions in a novel maze.

(A) An example LFP trace with ripples recorded in a subsequent session following the first session of novel exploration, where no ripple was detected.

(B) Ripple rates in the first and subsequent sessions recorded in the same animals (n=5 mice) during novel exploration.

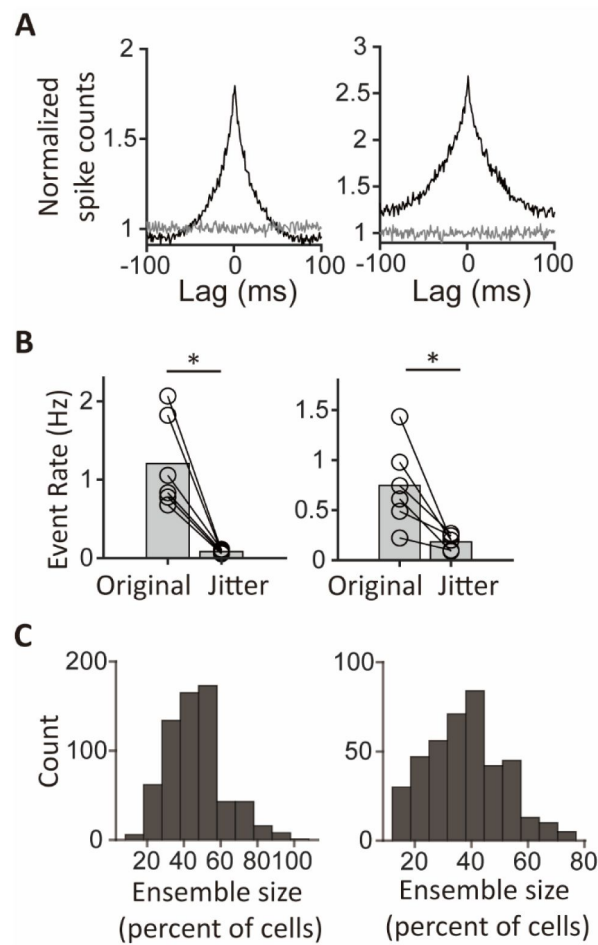


Figure 3-figure supplement 2.

Population synchrony during novel immobility.

(A) Grand average cross-correlograms (CCGs) for neurons labeled on E14.5 (left) and on E17.5 (right). The gray line represents the mean grand-average CCG between reference cells and randomly selected cells from different sessions.

(B) Pairwise comparison of synchronous event rates between the original and jittered data. Event rates were significantly higher in the original data compared to jittered controls for E14.5 neurons (left; mean event rate: 1.21 ± 0.24 Hz for the original data, 0.08 ± 0.009 Hz for jittered data, $n=6$ segments, $W=21$, $p=0.03$, Wilcoxon signed-rank test) and E17.5 neurons (right; mean event rate: 0.75 ± 0.17 Hz for the original data, 0.18 ± 0.03 Hz for jittered data, $n=6$ segments, $W=21$, $p=0.03$, Wilcoxon signed-rank test). Bar heights represent group means. * $p < 0.05$

(C) Histograms of ensemble sizes showing the percentage of cells participating in synchronous ensembles for E14.5 neurons (left) and E17.5 neurons (right).

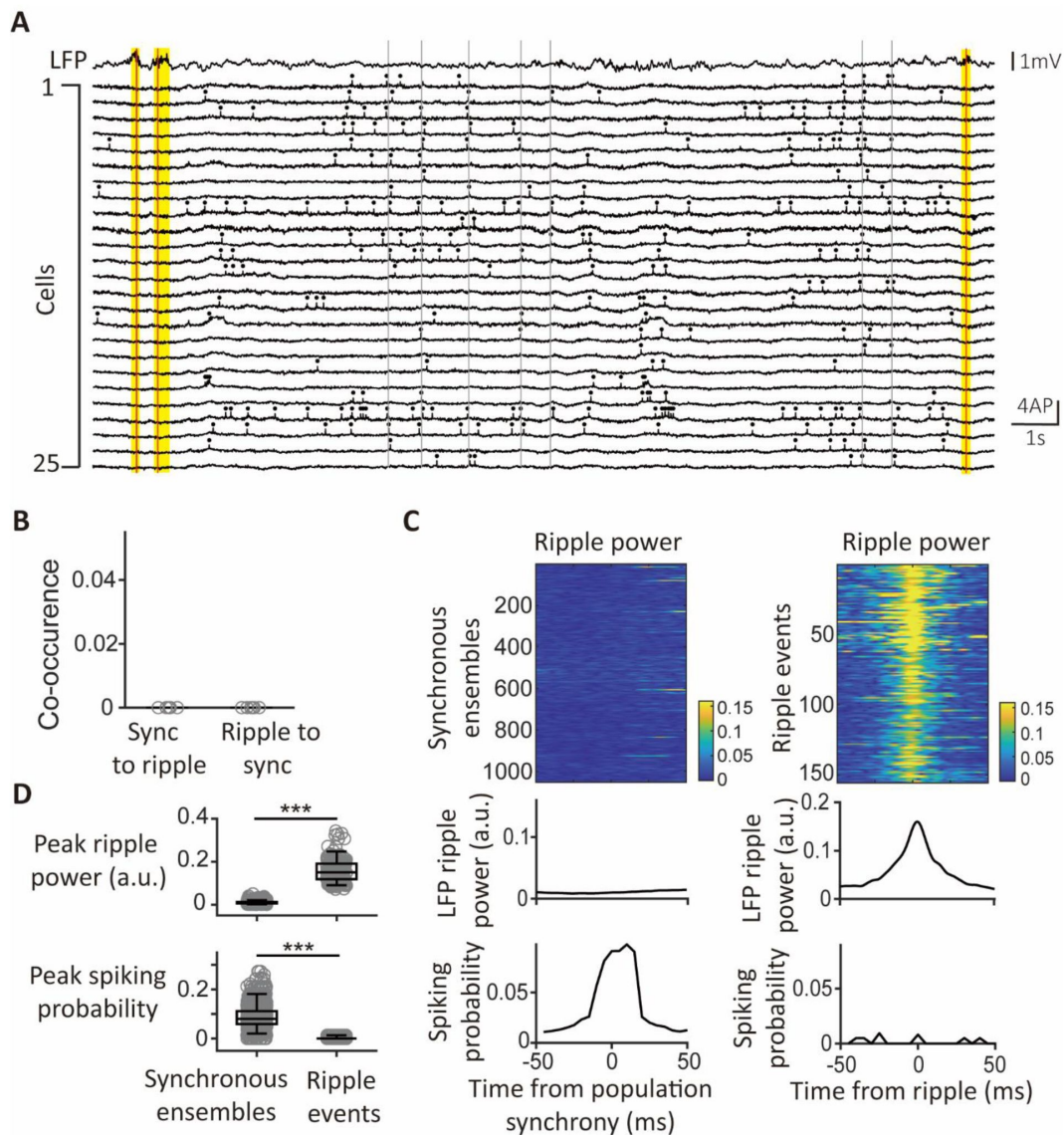


Figure 3-figure supplement 3.

Population synchrony outside ripple episodes during novel immobility.

(A) Example traces illustrating the distinct timings of ripple episodes and synchronous ensembles. Top: Simultaneously recorded LFP trace showing episodes of ripple oscillations. Red vertical lines indicate the timings of peak ripple power in the LFP trace, while yellow vertical bars mark the duration of ripple episodes. Bottom: Fluorescence traces from 25 CA1PCs. Gray vertical lines indicate the timings of synchronous ensembles.

(B) Percentages of co-occurrence between synchronous ensembles and ripple events. The “sync to ripples” group represents the percentage of ripple events that co-occur with synchronous ensembles, while the “ripples to sync” group shows the percentage of synchronous ensembles that co-occur with ripple events. Each gray circle represents an individual session.

(C) LFP ripple power and spiking probability aligned to synchronous ensembles and ripple events. **Upper panel:** Color-coded LFP power at ripple frequency (120-240Hz), aligned to the timings of 1052 synchronous ensembles (left) and 174 ripple events (right). **Middle panel:** Average LFP power at ripple frequencies aligned to the timings of synchronous ensembles (left) and ripple events (right). **Lower panel:** Average spiking probability of all recorded cells aligned to the timings of synchronous ensembles (left) and ripple events (right).

(D) Boxplots and comparisons of peak ripple power (upper panel) and peak spiking probability (lower panel) at the timings ($t=0$) of population synchrony (left) and ripple events (right). For peak ripple power: $p < 0.001$, $t_{(1224)} = 86.6$, Student’s t -test; for peak spiking probability: $p < 0.001$, $t_{(1224)} = 24.3$, Student’s t -test. Gray circles indicate peak values for individual triggered traces.

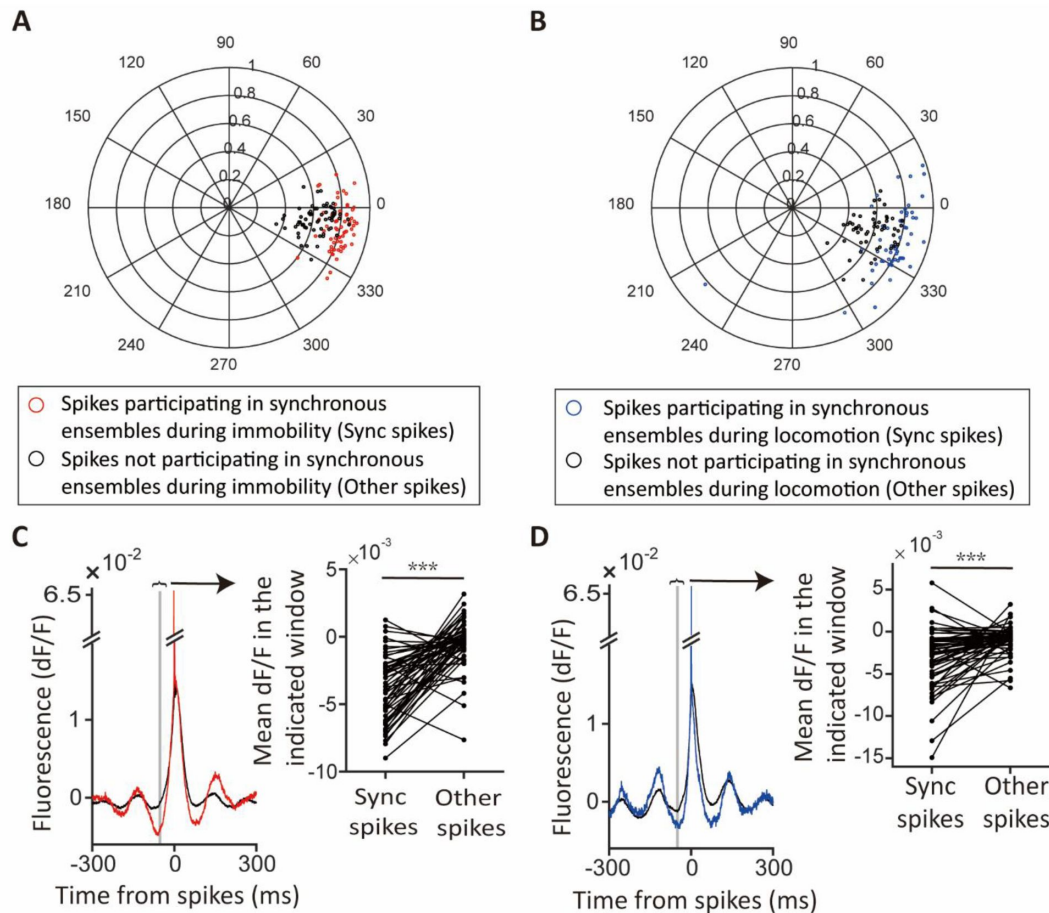


Figure 5-figure supplement 1.

SubVm theta modulation of spikes during immobility and locomotion periods of novel exploration.

(A) Polar plot comparing subVm theta modulation between spikes participating in synchronous ensembles (sync spikes) and spikes not participating in synchronous ensembles (other spikes) during immobility. Each dot represents the averaged modulation of a cell. Cells with modulation strengths that are not significant are excluded in the plot and in the comparison. The modulation strengths of spikes participating in synchrony are significantly higher than those of spikes not participating in synchrony (modulation strength for spikes participating in synchronous ensembles: 0.81 ± 0.07 , for spikes not participating in synchronous ensembles: 0.64 ± 0.11 , mean $\pm$ s.d., $t_{(64)} = 12.6$, $p < 0.001$, pair Student's t -test).

(B) Same as in (A), but during locomotion (modulation strength for spikes participating in synchronous ensembles: 0.80 ± 0.09 , for spikes not participating in synchronous ensembles: 0.60 ± 0.12 , mean $\pm$ s.d., $t_{(53)} = 12.3$, $p < 0.001$, pair Student's t -test).

(C) Average fluorescence triggered by cells' spikes participating in synchronous ensembles (red, sync spikes) and not participating in synchronous ensembles during immobility (black, other spikes; $n = 59$ cells). Inset: Pairwise comparison of average fluorescence 50 ± 5 ms before time of spikes ($t = 0$) between the two triggered groups ($n = 59$ cells, $p < 0.001$, pair Student's t -test). *** $p < 0.001$.

(D) Same as in (C), but during locomotion. Inset: Pairwise comparison of average fluorescence ($n = 59$ cells, $p < 0.001$, pair Student's t -test).

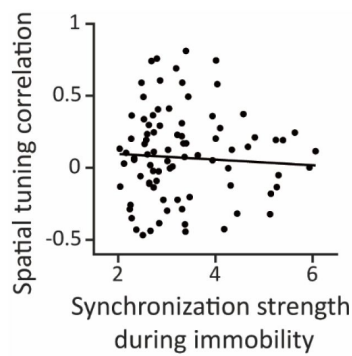
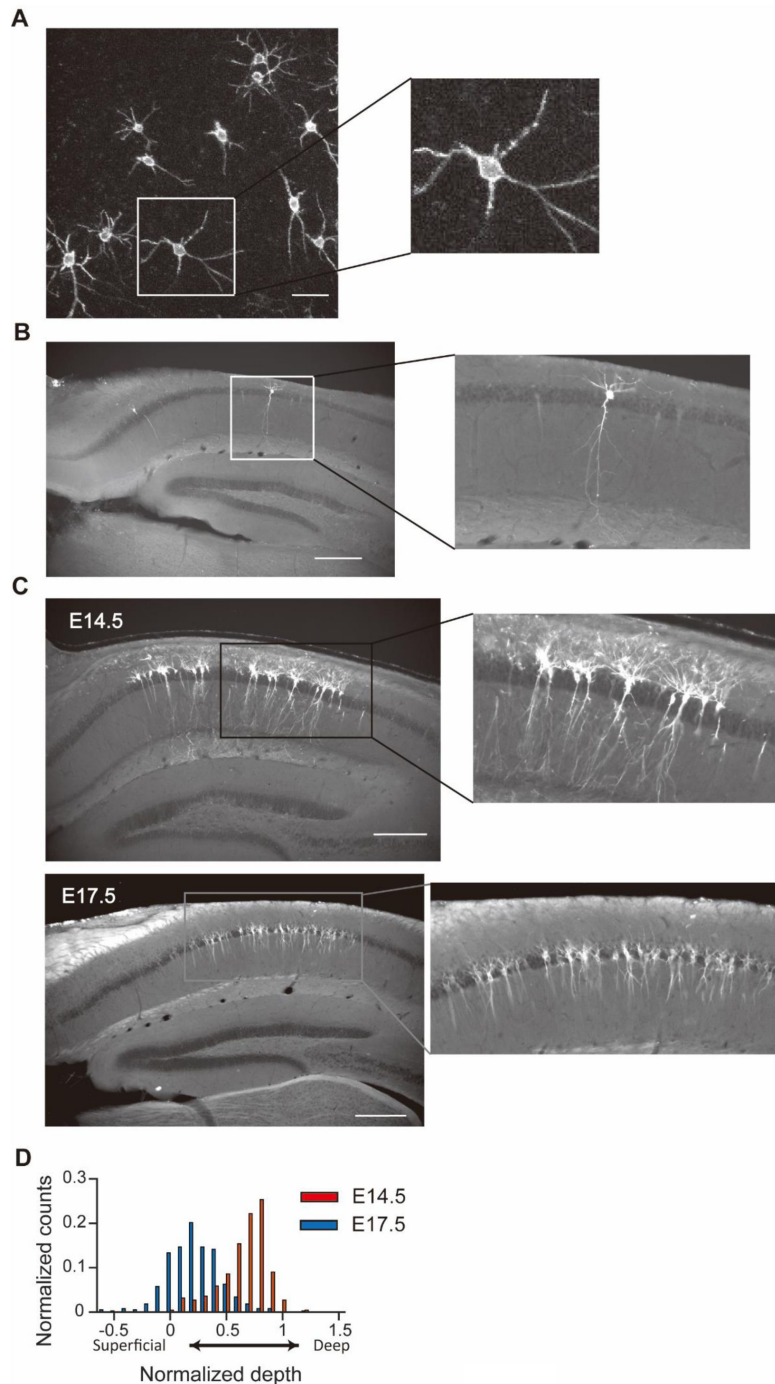


Figure 5-figure supplement 1.

Synchronization strengths during immobility show little correlation with the similarity of spatial tunings between place cells.

Synchronization strengths during immobility plotted against correlation coefficients of spatial tunings for pairs of place cells. Synchronization strengths show little correlation with similarities of spatial tunings (Spearman correlation coefficient=-0.003, $p=0.98$, $n=88$ cell pairs).



Supplementary Figure 1.

Morphology of Voltron-expressing neurons and their positions along the deep-superficial axis.

(A) Two-photon z-stack maximum intensity projection images of Voltron-expressing neurons obtained after imaging experiments. These neurons exhibit well-preserved membrane and dendritic morphology. Scale bars: 50 μ m.

(B) Post-hoc histological images of a sparsely labeled neuron previously subjected to voltage imaging. Scale bars: 300 μ m.

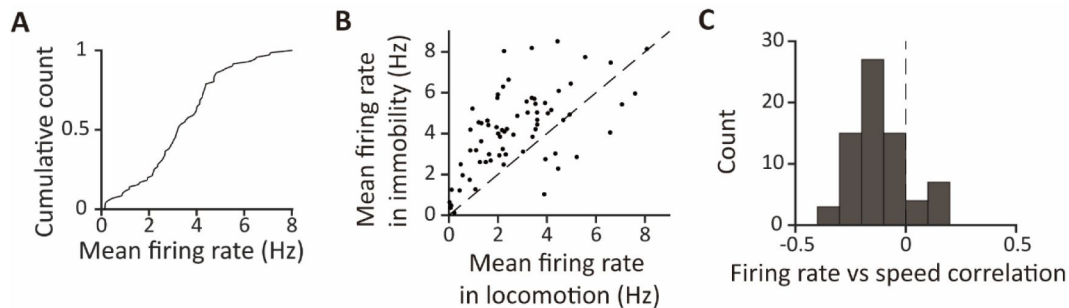
(C) Micrographs of labeled neurons from *in-utero* electroporation at embryonic day 14.5 (E14.5; upper panel) and E17.5 (lower panel). Scale bars: 300 μ m.

(D) Histograms of normalized depths for E14.5 neurons (red) and E17.5 neurons (blue). A higher normalized depth indicates that the soma is located deeper within the pyramidal layer. Normalized depths for E14.5 neurons are significantly higher than those for E17.5 neurons ($t_{(601)}=22.8$, $p<0.0001$, Student's *t*-test).

Supplementary Figure 2.

Mean firing rate distribution and correlation with animals' speed.

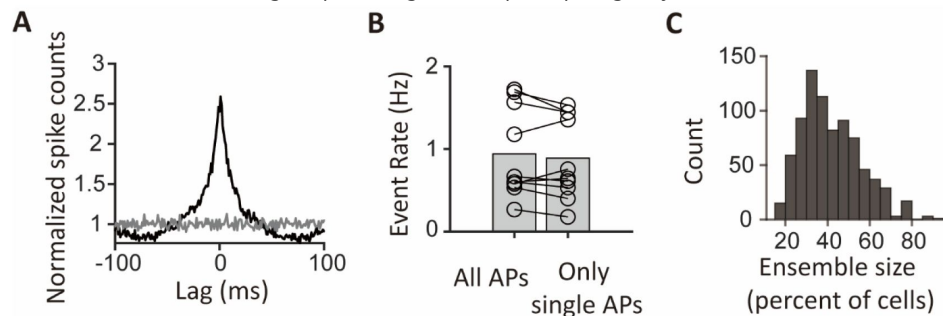
- (A) Cumulative distribution of mean firing rates averaged throughout the entire 180-second recording period.
 (B) Mean firing rates during locomotion plotted against mean firing rates during immobility for individual cells. Each dot represents a cell. Mean firing rates are higher during immobility than during locomotion (4.2 ± 0.2 Hz for immobility, 2.8 ± 0.2 Hz for locomotion, $t_{(70)} = 6.63$, $p < 0.0001$, Student's t -test).
 (C) Histogram of correlation coefficients between instantaneous firing rates and the animals' speed for each neuron.



Supplementary Figure 3.

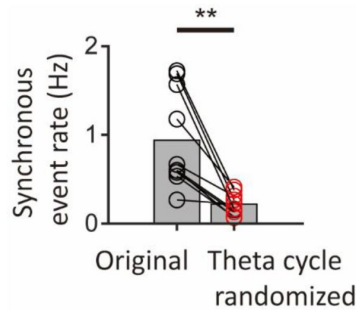
Population synchrony remains after removing the later spikes in bursts.

- (A) Grand-average cross-correlogram (CCG) calculated using spike trains with the later spikes in bursts removed. The gray line represents the mean grand-average CCG between reference cells and randomly selected cells from different sessions.
 (B) Pairwise comparison of population synchrony event rates between spike trains containing all spikes and those with the later spikes in bursts removed. Bar heights represent group means ($n = 10$ segments, $p = 0.36$, Wilcoxon signed-rank test).
 (C) Histogram of ensemble sizes showing the percentage of cells participating in synchronous ensembles.



Supplementary Figure 4.

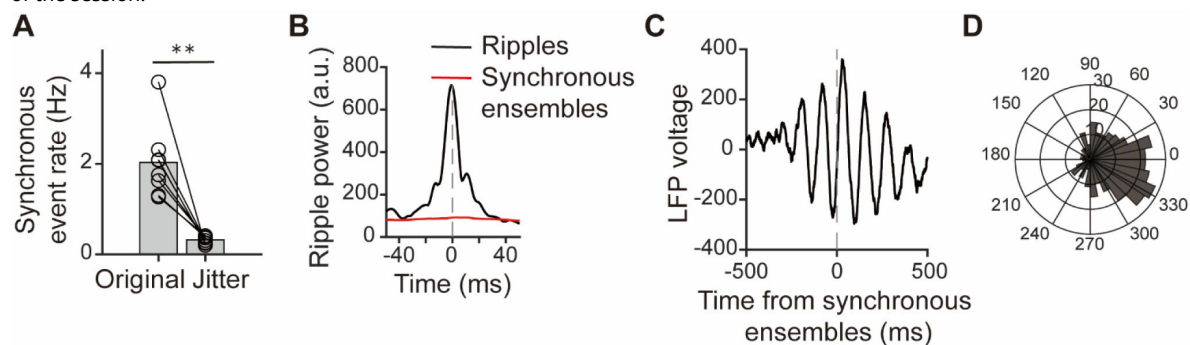
Significant reduction of population synchrony by randomizing spikes across theta cycles while preserving phases Pairwise comparison of population synchrony event rates between the original spike trains and randomized spike trains, in which spike timings were shuffled across theta cycles while preserving phases ($n=10$ segments, $p=0.002$, Wilcoxon signed-rank test). Bar heights represent group means. ** $p<0.01$



Supplementary Figure 5.

Analyses of synchronous ensembles in the publicly available data

- (A) Pairwise comparison of synchronous event rates between the original and jittered data. Event rates were significantly higher in the original data compared to jittered controls ($n=8$ sessions, $p=0.008$, Wilcoxon signed-rank test). Bar heights represent group means. ** $p<0.01$.
- (B) Dynamics of ripple power aligned to timings of ripples (black) and timings of synchronous ensembles (red).
- (C) Average LFP trace aligned to timings of synchronous ensembles of a session.
- (D) Polar histogram of the differences between theta phases of individual synchronous ensembles and the preferred phase of the session.



Acknowledgements

We express our sincere gratitude to Dr. Luke Lavis for sharing the JF₅₅₂-HaloTag ligand. Special thanks to Mr. Yu-Ting Lin and Ms. Hui-Ching Chen for technical assistance. We are grateful to Drs. Shih-Chieh Lin, Kuo-Hua Huang, Shi-Bing Yang, and Fu-Chin Liu for their insightful comments on the manuscript.

Additional information

Funding statements

National Science and Technology Council, Taiwan (B.-J.L. and T.-W.C.)

Howard Hughes Medical Institute, USA (E.R.S.)

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Author information

En-Li Chen

Institute of Neuroscience, National Yang Ming Chiao Tung University, Taipei, Taiwan, Brain Research Center, National Yang Ming Chiao Tung University, Taipei, Taiwan

Tsai-Wen Chen

Institute of Neuroscience, National Yang Ming Chiao Tung University, Taipei, Taiwan, Brain Research Center, National Yang Ming Chiao Tung University, Taipei, Taiwan

Eric R Schreiter

Janelia Research Campus, Howard Hughes Medical Institute, Ashburn, United States
ORCID iD: [0000-0002-2864-7469](https://orcid.org/0000-0002-2864-7469)

Bei-Jung Lin

Institute of Neuroscience, National Yang Ming Chiao Tung University, Taipei, Taiwan, Brain Research Center, National Yang Ming Chiao Tung University, Taipei, Taiwan
ORCID iD: [0000-0002-1755-1664](https://orcid.org/0000-0002-1755-1664)

For correspondence: beijunglin@nycu.edu.tw

Editors

Reviewing Editor

Laura Colgin

University of Texas at Austin, Austin, United States of America

Senior Editor

Laura Colgin

University of Texas at Austin, Austin, United States of America

Reviewer #1 (Public review):

Summary:

For many years, there has been extensive electrophysiological research investigating the relationship between local field potential patterns and individual cell spike patterns in the hippocampus. In this study, using state-of-the-art imaging techniques, they examined spike synchrony of hippocampal cells during locomotion and immobility states. In contrast to conventional understanding of the hippocampus, the authors demonstrated that hippocampal place cells exhibit prominent synchronous spikes locked to theta oscillations.

Strengths:

The voltage imaging used in this study is a highly novel method that allows recording not only suprathreshold-level spikes but also subthreshold-level activity. With its high frame rate, it offers time resolution comparable to electrophysiological recordings.

<https://doi.org/10.7554/eLife.96718.2.sa3>

Reviewer #2 (Public review):

Summary:

This study employed voltage imaging in the CA1 region of the mouse hippocampus during the exploration of a novel environment. The authors report synchronous activity, involving almost half of the imaged neurons, occurred during periods of immobility. These events did not correlate with SWRs, but instead, occurred during theta oscillations and were phased locked to the trough of theta. Moreover, pairs of neurons with high synchronization tended to display non-overlapping place fields, leading the authors to suggest these events may play a role in binding a distributed representation of the context.

Strengths:

Technically this is an impressive study, using an emerging approach that allow single-cell resolution voltage imaging in animals, that while head-fixed, can move through a real environment. The paper is written clearly and suggests novel observations about population-level activity in CA1.

Weaknesses:

The evidence provided is weak, with the authors making surprising population-level claims based on a very sparse data set (5 data sets, each with less than 20 neurons simultaneously recorded) acquired with exciting, but less tested technology. Further, while the authors link these observations to the novelty of the context, both in the title and text, they do not include data from subsequent visits to support this. Detailed comments are below:

- (1) My first question for the authors, which is not addressed in the discussion, is why these events have not been observed in the countless extracellular recording experiments conducted in rodent CA1 during exploration of novel environments. Those data sets often have 10x the neurons simultaneously recording compared to these present data, thus the highly synchronous firing should be very hard to miss. Ideally, the authors could confirm their claims via the analysis of publicly available electrophysiology data sets. Further, the claim of high extra-SWR synchrony is complicated by the observation that their recorded neurons fail to spike during the limited number of SWRs recorded during behavior- again, not agreeing with much of the previous electrophysiological recordings.
- (2) The authors posit that these events are linked to the novelty of the context, both in the

text, as well as in the title and abstract. However they do not include any imaging data from subsequent days to demonstrate the failure to see this synchrony in a familiar environment. If these data are available it would strengthen the proposed link to novelty if they were included.

(3) In the discussion the authors begin by speculating the theta present during these synchronous events may be slower type II or attentional theta. This can be supported by demonstrating a frequency shift in the theta recording during these events/immobility versus the theta recording during movement.

(4) The authors mention in the discussion that they image deep layer PCs in CA1, however this is not mentioned in the text or methods. They should include data, such as imaging of a slice of a brain post-recording with immunohistochemistry for a layer specific gene to support this.

Comments on revisions:

I have no further major requests and thank the authors for the additional data and analyses.

<https://doi.org/10.7554/eLife.96718.2.sa2>

Reviewer #3 (Public review):

Summary:

In the present manuscript, the authors use a few minutes of voltage imaging of CA1 pyramidal cells in head-fixed mice running on a track while local field potentials (LFPs) are recorded. The authors suggest that synchronous ensembles of neurons are differentially associated with different types of LFP patterns, theta and ripples. The experiments are flawed in that the LFP is not "local" but rather collected the other side of the brain.

Strengths:

The authors use a cutting-edge technique.

Weaknesses:

The two main messages of the manuscript indicated in the title are not supported by the data. The title gives two messages that relate to CA1 pyramidal neurons in behaving head-fixed mice: (1) synchronous ensembles are associated with theta (2) synchronous ensembles are not associated with ripples. The main problem with the work is that the theta and ripple signals were recorded using electrophysiology from the opposite hemisphere to the one in which the spiking was monitored. However, both rhythms exhibit profound differences as a function of location.

Theta phase changes with the precise location along the proximo-distal and dorso-ventral axes, and importantly, even reverses with depth. Because the LFP was recorded using a single-contact tungsten electrode, there is no way to know whether the electrode was exactly in the CA1 pyramidal cell layer, or in the CA1 oriens, CA1 radiatum, or perhaps even CA3 - which exhibits ripples and theta which are weakly correlated and in anti-phase with the CA1 rhythms, respectively. Thus, there is no way to know whether the theta phase used in the analysis is the phase of the local CA1 theta.

Although the occurrence of CA1 ripples is often correlated across parts of the hippocampus, ripples are inherently a locally-generated rhythm. Independent ripples occur within a fraction of a millimeter within the same hemisphere. Ripples are also very sensitive to the precise depth - 100 micrometers up or down, and only a positive deflection/sharp wave is

evident. Thus, even if the LFP was recorded from the center of the CA1 pyramidal layer in the contralateral hemisphere, it would not suffice for the claim made in the title.

<https://doi.org/10.7554/eLife.96718.2.sa1>

Author response:

The following is the authors' response to the original reviews.

We thank all reviewers for their thorough and thoughtful comments. We have carefully addressed each point raised, conducting new experiments and analyses to strengthen the manuscript. Below is a summary:

- Synchronous ensembles in new experiments: New experiments demonstrated synchronous ensembles during immobility in a novel environment (Figure 3-figure supplement 2) and revealed a significant reduction in such synchrony following familiarization training (Figure 4D).
- Ripple-associated activity: We detected a much larger number of ripple events to confirm (a) the suppression of CA1PC spiking during ripples (Figure 4Ai) and (b) that synchronous ensembles mostly occur outside ripples (Figure 3-figure supplement 3). Additionally, spiking suppression was accompanied by decreased subthreshold membrane potentials (Figure 4Bi, Ci). Ripple-associated spiking and membrane potential dynamics shifted toward higher firing rates and more depolarization after familiarization training (Figure 4).
- Public data analysis: Analysis of publicly available data identified theta-associated synchronous ensembles, demonstrating the generalizability of our findings across different experimental conditions (Supplementary Figure 5).
- Neuron morphology and algorithm validation: Images of recorded neurons after experiments confirmed their intact morphology. We also provided details on validating spike detection algorithms (Methods and Supplementary Figure 1).
- Cell soma locations: New data and analyses illustrate the distribution of cells labeled at different embryonic days along the radial axis of the pyramidal layer (Supplementary Figure 1).
- Analyses testing the robustness of synchronous ensembles: Additional analyses examined the impact of complex bursts and theta phase locking, confirming the robustness of synchronous ensembles detection (Supplementary Figures 3 and 4).
- Additional analyses and figures: We conducted further analyses and created new figures to address all remaining concerns (Response to Reviewer Figures 1-6).

We believe these revisions have significantly enhanced the paper, and we sincerely thank all reviewers for their invaluable feedback.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

For many years, there has been extensive electrophysiological research investigating the relationship between local field potential patterns and individual cell spike patterns in the hippocampus. In this study, using state-of-the-art imaging techniques, they examined spike synchrony of hippocampal cells during locomotion and immobility states. In

contrast to conventional understanding of the hippocampus, the authors demonstrated that hippocampal place cells exhibit prominent synchronous spikes locked to theta oscillations.

Strengths:

The voltage imaging used in this study is a highly novel method that allows recording not only suprathreshold-level spikes but also subthreshold-level activity. With its high frame rate, it offers time resolution comparable to electrophysiological recordings. Moreover, it enables the visualization of actual cell locations, allowing for the examination of spatial properties (e.g., Figure 4G).

We thank the reviewer for recognizing the strength of our study.

Weaknesses:

There is a notable deviation from several observations obtained through conventional electrophysiological recordings. Particularly, as mentioned below in detail, the considerable differences in baseline firing rates and no observations of ripple-triggered firing patterns raise some concerns about potential artifacts from imaging and analysis, such as cell toxicity, abnormal excitability, and false detection of spikes. While these findings are intriguing if the validity of these methods is properly proven, accepting the current results as new insights is challenging.

We appreciate the reviewer's insightful comments regarding the apparent deviation of our observation from conventional understanding, which we address in the following sections.

Reviewer #1 (Recommendations For The Authors):

(1) I am not particularly inclined to strongly adhere to conventional insights, but the findings obtained through this imaging method seem significantly different from those known from conventional electrophysiological recordings. For instance, there are noticeable differences in several basic firing characteristics. First, the average firing rates of 2.3-4.3 Hz (Line 97) appear higher than the distribution of firing frequencies reported in many electrophysiological recordings of pyramidal cells (e.g., Mizuseki et al., Cell Rep, 2013).

We understand that some of our findings differ from conventional insights. However, it is important to emphasize that many of our observations align closely with prior electrophysiological recordings. For instance, individual neurons in our study exhibit expected modulation by locomotion, spatial locations, novelty, and theta oscillations, all of which are hallmarks of normal hippocampal physiology.

Regarding the firing rates, it is important to highlight the heterogeneity of the firing rates, which range from 0.01 to 10 Hz, with a skewed distribution toward lower frequencies(1). While our values (2.3-4.3Hz) are higher than those reported by Mizuseki et al. (2013)(1) in rats, our recordings were obtained from mice and aligned with studies using mice, including firing rates of 2.1 Hz reported by McHugh et al. (1996) and 2.4-2.6 Hz by Buzsaki et al. (2003) (2,3).

In addition, our recordings were performed in a novel environment, which is known to enhance the firing rates of the hippocampal neurons(4). Consistent with this, our new recordings in a familiar environment demonstrate significantly lower firing rates (see below).

Results (line 279)

“Mean firing rates were significantly reduced in the familiar group compared to the novel group (Familiar group: 1.1 to 5.2 Hz (25th-75th percentiles), median=2.3 Hz, $n=66$ cells, 6 sessions, 4 mice; Novel group: 1.7 to 6.0 Hz (25th-75th percentiles), median=4.2 Hz, $n=111$ cells, 6 sessions, 6 mice, $p=0.0083$, Wilcoxon signed-rank test).”

Second, while this finding suggests that spike synchrony is entirely unrelated to ripple-triggered events, it is indeed difficult to comprehend (researchers who have analyzed electrophysiological data, at the very least, should have experienced some degree of correlation between ripples and spikes).

We thank the reviewer for raising this important point. We, too, found it surprising that population synchrony appears largely unrelated to ripples. To ensure the robustness of this observation, we conducted new experiments under conditions optimized for ripple detection to (a) confirm that the lack of positive correlation is also observed under conditions where we can detect more ripples and (b) demonstrate that our imaging methods can detect a higher correlation between ripples and spikes in a familiar environment (see details below).

Results (line 251)

“It was puzzling that these CA1PCs exhibited robust spiking activities outside of ripples yet generated few spikes during ripples. To further investigate neuronal activities during ripples, we established a recording condition that allowed us to capture more ripple episodes. Specifically, we immobilized mice in a tube to promote behaviors favoring ripple generation. The mice were habituated to head fixation in a tube in a room distinct from the one where imaging experiments were conducted. On the imaging day, the mice were introduced to the recording room and head-fixed under the microscope for the first time.

CA1PCs were labeled *in utero* on embryonic day (E) 14.5 ($n=56$ cells from 3 sessions in 3 mice) and E17.5 ($n=55$ cells from 3 sessions in 3 mice) and imaged in adult brains. Both neuronal populations exhibited prominent peaks in their grand average CCGs and significantly higher synchronous event rates compared to jittered data (Figure 3-figure supplement 2A, B). Approximately 40% of the recorded neurons participated in synchronous ensembles, indicating robust synchronous activity involving a substantial proportion of the recorded cells (Figure 3-figure supplement 2C).

In total, 1052 synchronous ensembles and 174 ripple episodes were detected across these imaging sessions. Consistent with findings from walking animals, few synchronous ensembles occurred during ripples when animals were immobilized in a tube (Figure 3-figure supplement 3A, B). Moreover, no distinguishable ripple oscillations were observed in synchronous events, and the average firing rates during ripple episodes were near zero (Figure 3-figure supplement 3C, D). At the single-cell level, 90% of neurons showed significant negative spiking modulation during ripples, with ripple modulation indexes close to -1, indicating strong suppression of spiking (Figure 4Ai). This suppression extended to subthreshold membrane potentials, as nearly all cells exhibited decreased fluorescence during ripples compared to baseline (Figure 4Bi, Ci). These results demonstrate that spiking activity and subthreshold membrane potentials are robustly suppressed during ripples.

Contextual novelty plays a critical role in shaping hippocampal neuronal activities. To assess its influence, we trained mice to become familiar with the imaging procedure and the recording environment over five days and recorded CA1PC activities on the final day. Mean firing rates were significantly reduced in the familiar group compared to the novel group (Familiar group:

1.1 to 5.2 Hz (25th-75th percentiles), median=2.3 Hz, $n=66$ cells, 6 sessions, 4 mice; Novel group: 1.7 to 6.0 Hz (25th-75th percentiles), median=4.2 Hz, $n=111$ cells, 6 sessions, 6 mice,

$p=0.0083$, Wilcoxon signed-rank test). Additionally, 15% of the neurons in the familiar group exhibited significantly positive spiking modulation by ripples, while fewer cells showed negative modulation compared to the novel group (Figure 4A). During ripples, neurons in the novel group predominantly displayed hyperpolarizing membrane voltage responses, whereas a subset of neurons in the familiar group exhibited prominent depolarizing responses (Figure 4B). The mean fluorescence changes in the familiar group shifted toward depolarization compared to the novel group (Figure 4C). Finally, synchronous event frequencies were significantly lower in the familiar context, indicating weaker synchronous activities under familiar conditions (Figure 4D). These results demonstrate that hippocampal neuronal activities, particularly synchronous ensembles, are strongly influenced by contextual novelty.”

Third, the fact that more than 40% of cells frequently exhibit synchronous firing other than during ripples has not been reported before, and if it were the case, many electrophysiologists would have likely noticed it. Overall, the excitability of cells seems too high.

We thank the reviewer for raising this point. As discussed above, the reported spike rates are within the range expected from the previous electrophysiology recordings in mice, especially given that we record cells in a novel environment. In addition, our jittering procedure ensures that the observed synchrony exceeds what could be expected from the given level of spike rates alone. These analyses support the robustness of our observations.

As mentioned below, there are concerns about experimental artifacts and analytical issues from optical imaging.

(2) Method: In surgery, the cortical tissue above the hippocampus was aspirated, which is a general method for in vivo calcium imaging from the hippocampus. Furthermore, they use a CAG promoter to express the sensors. To my knowledge, this promoter is excessively strong and may sometimes be toxic to cells. In addition, for imaging, they use DMSO and Pluronic F-127, which are relatively toxic materials (please describe their concentrations). These conditions might be damaging to hippocampal neurons.

We thank the reviewer for raising these comments. As the reviewer mentioned, cortical aspiration is a general method for *in vivo* imaging from the hippocampus and has been employed in numerous studies, including behavioral and systems-level investigations(5-15). For example, place cells are routinely recorded in both familiar and novel environments using this method and other approaches. Additionally, synchronous population activities have been observed and studied in the hippocampus both with and without cortical aspiration(6,15-18). These findings demonstrate that the hippocampal neuronal network generates place cells and synchronous activities regardless of whether the cortical tissue above it has been aspirated.

DMSO and Pluronic F-127 are used as solvents for dissolving the JF₅₅₂HaloTag ligand, and the resulting solution is injected into the bloodstream rather than directly into brain tissue. The concentrations of these reagents in the dye solution are now described in the text (see below). Assuming a blood volume of 2 ml in adult mice, the final concentrations of DMSO and Pluronic F-127 in the bloodstream are estimated to be 1% upon injection and then decrease rapidly while they are metabolized and excreted out of the body. Moreover, the effective concentrations in the brain tissue would be even lower. These low concentrations have been demonstrated to have minimal impact on cells and tissue(19-22).

Methods (line 616)

“JF₅₅₂-HaloTag ligand (a generous gift from Dr. Luke Lavis) was first dissolved in DMSO (20 μ l, Sigma) and then diluted in PluronicTM F-127 (20 μ l, P3000MP, Invitrogen) and PBS to achieve a final concentration of 0.83 mM of JF₅₅₂-HaloTag ligand. The solution was then injected intravenously through the retro-orbital sinus. Imaging sessions were initiated 3 hours after the injection of the JF₅₅₂-HaloTag ligand.”

We understand that the CAG promoter may sometimes be toxic to cells if it drives high expression. However, it is important to note that we injected highly diluted virus (20x, final titer: 2.7×10^{12} GC/ml) to avoid excessive expression levels. This titer was determined from serial dilution experiments to ensure an optimal expression level free from toxicity (see below). The same titer was used in a previous study(23) to label CA1 interneurons, which exhibited physiological spike rates and synchrony (see Abdelfattah 2023, Neuron, Figure 8). Furthermore, Voltron expression does not significantly affect key cellular properties, including membrane resistance, membrane capacitance, resting membrane potentials, spike amplitudes, and spike width (see Abdelfattah 2019, Science, Supplementary Figures 11 and 12). In our recordings, individual neurons exhibit the expected modulation by locomotion, spatial locations, novelty, and theta oscillations. We now include images of the recorded neurons to demonstrate their intact morphology and healthy appearance following imaging experiments (Supplementary Figure 1A, B), further supporting minimal cytotoxic effects.

Methods (line 577)

“A serial dilution experiment was conducted to determine an optimal titer of the virus carrying Voltron2 genes, minimizing cell toxicity, for use in this and in previous imaging experiments. A fine injection pipette (tip diameter 10-60 μ m) was used to inject AAV2/1-CAG-flex-Voltron2-ST (2.7×10^{12} GC/ml, a generous gift from Dr. Eric Schreiter and the GENIE team at HHMI Janelia Research Campus) into the exposed regions at a depth of 200 μ m (up to six injection sites and 100-200 nL of viral suspension).”

(3) Another concern is the relatively low number of cells simultaneously recorded during imaging compared to typical hippocampal imaging such as Inscopix which often records several hundred cells. In this study, however, this number is 20 or fewer. This is likely because the visualized cells at baseline were limited to this extent. It is possible that these cells represent particularly too strong sensor expression, which may facilitate visualization and high signal-to-noise ratio in voltage imaging. Consequently, there is a possibility of abnormal activity occurring in these cells.

The Inscopix studies use calcium imaging, which has a temporal resolution that is too slow to resolve fast synchrony central to our study. To enable highspeed voltage imaging at 2000 frames per second, we employed strategies to achieve sparse labeling and carefully limited the number of labeled cells to minimize out-of-focus contamination. In our analysis, we applied a criterion to include only cells separated by 70 μ m or longer, reducing the potential for channel cross-talk among nearby neurons. These criteria limited the number of simultaneously imaged cells in our experiments. To address this issue, we have now included new data from 12 additional animals with 177 neurons to support our findings.

Furthermore, despite the limited number of simultaneously imaged cells, population synchrony beyond what could be expected by chance can be detected using rigorous statistical procedures. As discussed earlier, neuronal activities were within the expected range; they were modulated by animals' locomotion (Figure 2 and Supplementary Figure 2), exhibited place tuning, and were significantly reduced when the recording context became familiar, supporting the normal physiology of the recorded cells.

(4) Analysis: There are some criteria for detecting spikes (described in the Methods), but there are concerns about whether these criteria truly extract only spike activity. When

examining the traces in Figure 1 and Figure 2, there appear to be some activities that show fluorescence increases up to the level of putative spikes. How can we determine that these are indeed subthreshold changes? Conversely, some activities detected as spikes may also be subthreshold synaptic potential (this possibility concerns me). There is a need for more precise validation of spike detection analysis to ensure its accuracy.

Regarding spike detection, we used validated algorithms(23-25) to ensure robust and reliable spike identification. Spiking activity was first separated from slower subthreshold potentials using high-pass filtering. This approach prevents slow fluorescence increases from being misinterpreted as spikes, even if their amplitude is large. We benchmarked this detection algorithm in our recent publication (Huang et al., 2024)(24), demonstrating its high sensitivity and specificity in spike detection (see the figure below). While we acknowledge that a small number of spikes, particularly those occurring later in a burst, might be missed due to their smaller amplitudes (as illustrated in Figures 1 and 2 of the manuscript), we anticipate that any missed spikes would lead to a decrease, rather than an increase, in synchrony between neurons. Overall, we are confident that spike detection is performed in a rigorous and reliable manner.

Method (line 670)

“Previous studies have described and validated the procedure for imaging preprocessing and spike detection. In short, the fluorescence intensities of individual neurons were calculated by averaging the fluorescence intensities of pixels from the same ROIs. Bleaching was corrected by calculating the baseline fluorescence (F_0) at each time point as an average of the fluorescence intensities within ± 0.5 seconds around the time point. The dF/F was calculated as the F_0 minus the fluorescence intensity of the same time point divided by F_0 . Positive fluorescence transients were detected to identify spikes from the high-passed dF/F traces created by subtracting the dF/F traces from the median-filtered version with a 5-ms window. To simulate the noise of recordings, high-passed dF/F traces were inverted, and the amplitudes of the transients detected from the inverted traces were used to construct a noise distribution of the spike amplitudes. A threshold was set by comparing the amplitudes of the detected transients with the noise distribution of the spike amplitudes to minimize the sum of type I and type II errors. Spikes were first detected when transients were larger than the threshold. Then, spike amplitudes smaller than half of the top 5% spike amplitudes were excluded. The signal-to-noise ratio (SNR) was calculated for each neuron as a ratio of the averaged spike amplitudes over the standard deviation of the high-passed dF/F traces, excluding points 2 ms before and 4 ms after each detected spike to estimate the quality of the recordings.”

(5) If the authors aim to establish this new physiological phenomenon, it is necessary to compare it with electrophysiological data or verify if similar phenomena can be detected from electrophysiological data. Recently, various datasets have been made publicly available (e.g. CRCNS and Mendeley data), and these should be easily verifiable without the need for conducting experiments.

We thank the reviewer for the suggestion. To address this, we analyzed a publicly available dataset (hc-11 on CRCNS), which contains hippocampal recordings from rats navigating novel mazes for water rewards. Using our algorithm, we detected significant population synchrony in the dataset (Supplementary Figure 5A). The synchronous event rates were 6.4-fold higher than those in jittered controls, demonstrating the reliability of our findings.

Additionally, these synchronous events mostly occurred in the absence of ripples and were coupled to theta oscillations (Supplementary Figure 5B-D). These results not only validate our findings using independent datasets but also highlight the generalizability of synchronous ensembles as a distinct network phenomenon relevant to hippocampal function.

Results (line 366)

“To further investigate synchronous ensembles across different datasets, we analyzed publicly available hippocampal recordings ‘hc-11’ from the CRCNS repository, where rats navigated novel mazes for water rewards (see Method). Using our algorithm, we identified a significant number of synchronous ensembles during the first three minutes of novel navigation. On average, the rates of synchronous events were 6.4-fold higher than those detected in jittered controls (mean event rate: 2.0 ± 0.3 Hz for the original data vs. 0.32 ± 0.03 Hz for jittered data, $n = 8$ sessions, $p = 0.0078$, $W = 36$, Wilcoxon signedrank test; Supplementary Figure 5A). To assess whether ripple oscillations were associated with these synchronous ensembles, we analyzed ripple event rates and their relationship to population synchrony. During this period, ripple events were infrequent (mean ripple rate: 0.02 ± 0.01 , $n = 8$ sessions), and ripple power peaked during ripple episodes but remained low at the timings of population synchrony (Supplementary Figure 5B). Nevertheless, LFP traces aligned to population synchrony revealed prominent theta oscillations (Supplementary Figure 5C). Synchronous ensembles were modulated by LFP theta oscillation (modulation strength: 0.30 ± 0.04 , $n = 8$ sessions, $p < 0.001$), and the timings of individual ensembles were consistently locked to the preferred phase of each session, suggesting a functional coupling of synchronous ensembles to theta oscillations important for information processing (Supplementary Figure 5D).”

(6) Please describe exact statistical information (e.g. statistical values, degree of freedom, and test types) throughout the manuscript.

Statistical values, degree of freedom and test types have been included in the manuscript. Please see below an example in the manuscript:

Result (line 96)

“Consistent with previous studies, neurons labeled on E14.5 located more on the deep side of the pyramidal layer than those labeled on E17.5 ($t_{(601)} = 22.8$, $p < 0.0001$, Student’s t -test; Supplementary Figure 1C, D).”

Minor comment - Figure 2A legend: what is "gray rectangles"?

We apologize for the inconsistency in nomenclature in the figure legends. We have now corrected this issue and consistently use the term “gray vertical bars” to indicate the timings and durations of synchronous events throughout the article.

Reviewer #2 (Public Review):

Summary:

This study employed voltage imaging in the CA1 region of the mouse hippocampus during the exploration of a novel environment. The authors report synchronous activity, involving almost half of the imaged neurons, occurred during periods of immobility. These events did not correlate with SWRs, but instead, occurred during theta oscillations and were phasedlocked to the trough of theta. Moreover, pairs of neurons with high synchronization tended to display non-overlapping place fields, leading the authors to suggest these events may play a role in binding a distributed representation of the context.

We thank the reviewer for a thorough and thoughtful review of our paper.

Strengths:

Technically this is an impressive study, using an emerging approach that allows single-cell resolution voltage imaging in animals, that while head-fixed, can move through a real environment. The paper is written clearly and suggests novel observations about population-level activity in CA1.

We thank the reviewer for pointing out the technical strength and the novelty of our study.

Weaknesses:

The evidence provided is weak, with the authors making surprising population-level claims based on a very sparse data set (5 data sets, each with less than 20 neurons simultaneously recorded) acquired with exciting, but less tested technology. Further, while the authors link these observations to the novelty of the context, both in the title and text, they do not include data from subsequent visits to support this. Detailed comments are below:

We understand the reviewer's concerns regarding the dataset size. In the revised manuscript, we have included additional data to further strengthen our conclusions and provide a more robust dataset. Specifically, we expanded our analysis by increasing the number of sessions and neurons recorded, ensuring that the findings are more representative and less likely to be influenced by sample sizes.

Moreover, synchronous ensembles exceeding what could be expected by chance were detected in all examined data, validating our claims regarding population synchrony. We have also carefully considered the potential impact of the technology used in our experiments and included additional validation and comparison with results from other studies employing complementary techniques to support the reliability of our conclusions.

Regarding the link to novelty, we have included data from subsequent visits, as suggested by the reviewer. These new data demonstrate that the observed changes in synchronous ensembles are context-dependent and significantly influenced by novelty. This confirms the novelty-related effects observed during initial visits and further supports the conclusions drawn in the manuscript. Please see below for our detailed replies to each of the reviewer's points.

(1) My first question for the authors, which is not addressed in the discussion, is why these events have not been observed in the countless extracellular recording experiments conducted in rodent CA1 during the exploration of novel environments. Those data sets often have 10x the neurons simultaneously recording compared to these present data, thus the highly synchronous firing should be very hard to miss. Ideally, the authors could confirm their claims via the analysis of publicly available electrophysiology data sets. Further, the claim of high extra-SWR synchrony is complicated by the observation that their recorded neurons fail to spike during the limited number of SWRs recorded during behavior- again, not agreeing with much of the previous electrophysiological recordings.

We thank the reviewer for raising these important questions. To address the first question, it is possible that synchronous ensembles were not previously detected in extracellular recordings due to differences in detection methods or analysis approaches. To investigate this further, we analyzed a publicly available dataset (hc-11 on CRCNs), which contains hippocampal recordings from rats navigating novel mazes for water rewards. Using our algorithm, we detected robust synchronous ensembles in the dataset (Supplementary Figure 5). The rates of synchronous events were significantly higher than those in jittered controls, demonstrating the reliability and generalizability of these synchronous ensembles.

Results (line 366)

“To further investigate synchronous ensembles across different datasets, we analyzed publicly available hippocampal recordings ‘hc-11’ from the CRCNS repository, where rats navigated novel mazes for water rewards (see Method). Using our algorithm, we identified a significant number of synchronous ensembles during the first three minutes of novel navigation. On average, the rates of synchronous events were 6.4-fold higher than those detected in jittered controls (mean event rate: 2.0 ± 0.3 Hz for the original data vs. 0.32 ± 0.03 Hz for jittered data, $n = 8$ sessions, $p = 0.0078$, $W = 36$, Wilcoxon signedrank test; Supplementary Figure 5A). To assess whether ripple oscillations were associated with these synchronous ensembles, we analyzed ripple event rates and their relationship to population synchrony. During this period, ripple events were infrequent (mean ripple rate: 0.02 ± 0.01 , $n = 8$ sessions), and ripple power peaked during ripple episodes but remained low at the timings of population synchrony (Supplementary Figure 5B). Nevertheless, LFP traces aligned to population synchrony revealed prominent theta oscillations (Supplementary Figure 5C). Synchronous ensembles were modulated by LFP theta oscillation (modulation strength: 0.30 ± 0.04 , $n = 8$ sessions, $p < 0.001$), and the timings of individual ensembles were consistently locked to the preferred phase of each session, suggesting a functional coupling of synchronous ensembles to theta oscillations important for information processing (Supplementary Figure 5D).”

To address the second question, we conducted new experiments under conditions optimized for ripple generation. Specifically, we recorded neurons in mice head-fixed in a novel environment, resulting in 174 ripple episodes across six sessions. Consistent with our original findings, spiking rates were significantly suppressed and membrane potentials were hyperpolarized during ripples (Figure 4Ai-Ci of the manuscript). Despite this suppression, the same neurons exhibit rich synchronous activities outside of ripples (Figure 3-figure supplement 3 of the manuscript). These results confirm that these synchronous ensembles are distinct from ripple-related neuronal activity and strengthen our claim that the observed synchronous ensembles represent a distinct physiological phenomenon, consistent across different datasets and experimental conditions.

Results (line 251)

“It was puzzling that these CA1PCs exhibited robust spiking activities outside of ripples yet generated few spikes during ripples. To further investigate neuronal activities during ripples, we established a recording condition that allowed us to capture more ripple episodes. Specifically, we immobilized mice in a tube to promote behaviors favoring ripple generation. The mice were habituated to head fixation in a tube in a room distinct from the one where imaging experiments were conducted. On the imaging day, the mice were introduced to the recording room and head-fixed under the microscope for the first time.

CA1PCs were labeled *in utero* on embryonic day (E) 14.5 ($n=56$ cells from 3 sessions in 3 mice) and E17.5 ($n=55$ cells from 3 sessions in 3 mice) and imaged in adult brains. Both neuronal populations exhibited prominent peaks in their grand average CCGs and significantly higher synchronous event rates compared to jittered data (Figure 3-figure supplement 2A, B). Approximately 40% of the recorded neurons participated in synchronous ensembles, indicating robust synchronous activity involving a substantial proportion of the recorded cells (Figure 3-figure supplement 2C).

In total, 1052 synchronous ensembles and 174 ripple episodes were detected across these imaging sessions. Consistent with findings from walking animals, few synchronous ensembles occurred during ripples when animals were immobilized in a tube (Figure 3-figure supplement 3A, B). Moreover, no distinguishable ripple oscillations were observed in synchronous events, and the average firing rates during ripple episodes were near zero (Figure 3-figure supplement 3C, D). At the single-cell level, 90% of neurons showed significant negative spiking modulation during ripples, with ripple modulation indexes close to -1,

indicating strong suppression of spiking (Figure 4Ai). This suppression extended to subthreshold membrane potentials, as nearly all cells exhibited decreased fluorescence during ripples compared to baseline (Figure 4Bi, Ci). These results demonstrate that spiking activity and subthreshold membrane potentials are robustly suppressed during ripples.”

(2) The authors posit that these events are linked to the novelty of the context, both in the text, as well as in the title and abstract. However, they do not include any imaging data from subsequent days to demonstrate the failure to see this synchrony in a familiar environment. If these data are available it would strengthen the proposed link to novelty if they were included.

Following the reviewer’s suggestion, we record neuronal activities in a familiar context to test the proposed link between synchronous activity and contextual novelty. We found that synchronous activity levels were significantly lower in the familiar context compared to the novel context, demonstrating that synchronous activity is strongly modulated by contextual novelty (Figure 4D of the manuscript). These findings provide further support for a link of the synchronous ensembles to novel environmental contexts.

Result (line 277)

“Contextual novelty plays a critical role in shaping hippocampal neuronal activities. To assess its influence, we trained mice to become familiar with the imaging procedure and the recording environment over five days and recorded CA1PC activities on the final day. Mean firing rates were significantly reduced in the familiar group compared to the novel group (Familiar group:

1.1 to 5.2 Hz (25th-75th percentiles), median=2.3 Hz, $n=66$ cells, 6 sessions, 4 mice; Novel group: 1.7 to 6.0 Hz (25th-75th percentiles), median=4.2 Hz, $n=111$ cells, 6 sessions, 6 mice, $p=0.0083$, Wilcoxon signed-rank test). Additionally, 15% of the neurons in the familiar group exhibited significantly positive spiking modulation by ripples, while fewer cells showed negative modulation compared to the novel group (Figure 4A). During ripples, neurons in the novel group predominantly displayed hyperpolarizing membrane voltage responses, whereas a subset of neurons in the familiar group exhibited prominent depolarizing responses (Figure 4B). The mean fluorescence changes in the familiar group shifted toward depolarization compared to the novel group (Figure 4C). Finally, synchronous event frequencies were significantly lower in the familiar context, indicating weaker synchronous activities under familiar conditions (Figure 4D). These results demonstrate that hippocampal neuronal activities, particularly synchronous ensembles, are strongly influenced by contextual novelty.”

(3) In the discussion the authors begin by speculating the theta present during these synchronous events may be slower type II or attentional theta. This can be supported by demonstrating a frequency shift in the theta recording during these events/immobility versus the theta recording during movement.

We thank the reviewer for the suggestion. As the reviewer points out, we did observe a frequency shift in synchrony-associated theta during immobility compared to locomotion (see Figure 5B, red vs. blue curves). We have now highlighted this result in the discussion section. Please refer to the text below.

Discussion (line 471)

“On the other hand, type 2 theta, or attentional theta, is slightly slower and is blocked by muscarinic receptor antagonists, emerging during states of arousal or attention, such as when entering a new environment. Consistent with these distinctions, the peak of the power

spectrum density shows a distinctively slower theta frequency during immobility compared to locomotion (Figure 5B)."

(4) The authors mention in the discussion that they image deep-layer PCs in CA1, however, this is not mentioned in the text or methods. They should include data, such as imaging of a slice of a brain post-recording with immunohistochemistry for a layer-specific gene to support this.

We thank the reviewer for the constructive suggestion. In response, we have added images of slices from both E14.5 and E17.5 brains and analyzed soma locations along the radial axis of the pyramidal layer. The results are included in the main text, Methods, and Supplementary Figure 1 of the manuscript (see below).

Result (line 96)

"Consistent with previous studies, neurons labeled on E14.5 located more on the deep side of the pyramidal layer than those labeled on E17.5 ($t_{(601)}=22.8$, $p<0.0001$, Student's t -test; Supplementary Figure 1C, D)."

Methods (line 563)

"The injection resulted in Cre expression among neurons born on the day of injection, with earlier injection labeling neurons located on the deeper side of the cell layer."

Reviewer #3 (Public Review):

Summary:

In the present manuscript, the authors use a few minutes of voltage imaging of CA1 pyramidal cells in head-fixed mice running on a track while local field potentials (LFPs) are recorded. The authors suggest that synchronous ensembles of neurons are differentially associated with different types of LFP patterns, theta and ripples. The experiments are flawed in that the LFP is not "local" but rather collected in the other side of the brain, and the investigation is flawed due to multiple problems with the point process analyses. The synchrony terminology refers to dozens of milliseconds as opposed to the millisecond timescale referred to in prior work, and the interpretations do not take into account theta phase locking as a simple alternative explanation.

We appreciate the reviewer's feedback and acknowledge the concerns raised. However, we believe these concerns can be effectively addressed without compromising the validity of our conclusions. With this in mind, we respectfully disagree with the assessment that our experiments and investigation are flawed. Please allow us to address these concerns and offer additional context to support the validity of our study.

Weaknesses:

The two main messages of the manuscript indicated in the title are not supported by the data. The title gives two messages that relate to CA1 pyramidal neurons in behaving head-fixed mice: (1) synchronous ensembles are associated with theta (2) synchronous ensembles are not associated with ripples.

There are two main methodological problems with the work: (1) experimentally, the theta and ripple signals were recorded using electrophysiology from the opposite hemisphere to the one in which the spiking was monitored. However, both signals exhibit profound differences as a function of location: theta phase changes with the precise location along the proximo-distal and dorso-ventral axes, and importantly, even reverses with depth. And ripples are often a local phenomenon - independent ripples occur within a fraction

of a millimeter within the same hemisphere, let alone different hemispheres. Ripples are very sensitive to the precise depth - 100 micrometers up or down, and only a positive deflection/sharp wave is evident.

We acknowledge the reviewer's consideration regarding the collection of LFP from the contralateral hemisphere. While we acknowledge the limitation of this design, we believe these contralateral LFP recordings still provide valuable insights into the dynamics of synchronous ensembles. Despite potential variations in theta phases due to differences in recording locations and depths, the occurrence and amplitudes of theta oscillations are generally wellcoordinated across hemispheres (Buzsaki et al., 2003, Fig 5)(3). The presence of prominent contralateral LFP theta activity around the times of synchronous ensembles in our study (Figure 5A of the manuscript) strongly supports our conclusion about their association with theta oscillations, even with LFP collected from the opposite hemisphere.

Additionally, we explicitly noted in the manuscript that the "preferred phases" varied between sessions, likely reflecting variability in recording locations (see below). Thus, we believe the concern about theta phase variability has already been adequately addressed in the current manuscript.

Result (line 321)

"Although the preferred phases varied from session to session due to differences in recording sites along the proximal-distal axis of the hippocampus, the timings of individual ensembles were consistently locked to the preferred phase of each session (Figure 5C)."

While we acknowledge that ripple oscillations can sometimes occur locally, the majority of ripples occur synchronously in both hemispheres (up to 70%)(3,26), as demonstrated both in the literature (Szabo et al., 2022, Supplementary Figure 2) and by data from our lab (Huang et al., 2024, Figure S6). As a result, using contralateral LFP to infer ripple occurrence on the ipsilateral side is a well-established practice in the field, commonly employed by many studies published in reputable journals(26-29). Given the high co-occurrence of both theta and ripple oscillations across hemispheres, we maintain that the two main messages of our manuscript are supported by data, despite the concern regarding phase discrepancy mentioned by the reviewer.

(2) The analysis of the point process data (spike trains) is entirely flawed. There are many technical issues: complex spikes ("bursts") are not accounted for; differences in spike counts between the various conditions ("locomotion" and "immobility") are not accounted for; the pooling of multiple CCGs assumes independence, whereas even conditional independence cannot be assumed; etc.

We acknowledge the reviewer's concern regarding spike train analysis. Complex bursts or differences in behavioral conditions can indeed lead to variations in spike counts, which could potentially affect the detection of synchronous ensembles. However, our jittering procedure is specifically designed to account for variations in spike counts. Notably, while the jittered spike trains retain the same spike count variations, we observed 7.8 times more synchronous events in our data compared to the jitter controls (Figure 1G of the manuscript). This indicates that the specific spike timings in the original data - disrupted in the jitter data - are responsible for the observed synchrony.

To further address the concern that complex bursts might influence the observed synchrony, we performed additional analyses in which we excluded all later spikes in bursts, considering only single spikes and the first spikes of bursts. Importantly, this procedure did not affect the rate or size of synchronous ensembles and did not significantly alter the grand-average CCG (Supplementary Figure 3). These results explicitly demonstrate that complex bursts do not significantly impact the analysis of synchronous ensembles.

Result (line 131)

The observed population synchrony was not attributable to spikes in complex bursts, as synchronous event rates did not differ significantly with or without the inclusion of later spikes in bursts (Supplementary Figure 3).

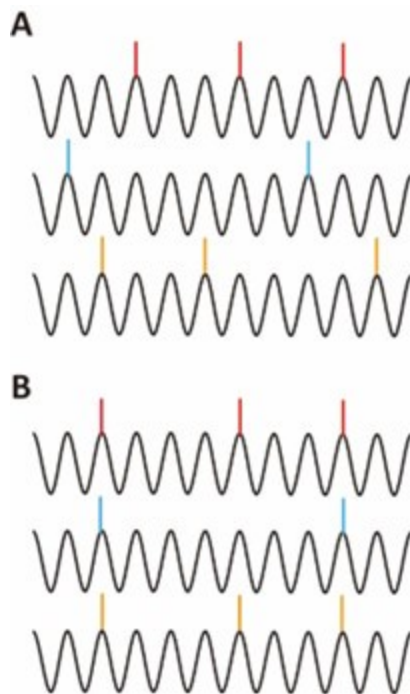
Beyond those methodological issues, there are two main interpretational problems: (1) the "synchronous ensembles" may be completely consistent with phase locking to the intracellular theta (as even shown by the authors themselves in some of the supplementary figures).

We agree with the reviewer that the synchronous ensembles are indeed consistent with theta phase locking. However, it is important to note that theta phase locking alone does not necessarily imply population synchrony. In fact, previous research has demonstrated that theta phase locking can "reduce" population synchrony(30). Thus, the presence of theta phase locking cannot be considered a simple alternative explanation for the synchronous ensembles.

The idea that theta phase locking does not necessarily lead to population synchrony is illustrated in Author response image 1A. In this example, while all three neurons are perfectly locked to specific theta phases, no synchrony among neurons is evident. In contrast, our data align with the scenario depicted in Figure 4B, where spikes occur not only at specific theta phases but also in the same cycles, thereby facilitating population synchrony.

Author response image 1.

Illustrative diagram of the relationship between theta phase coupling and population synchrony. Illustration of theta phase coupling with low population synchrony. Illustration of population synchrony with theta phase coupling.



To directly assess the contribution of theta phase locking to synchronous ensembles, we performed a new analysis in which the specific theta cycles during which neurons spike were

randomized while keeping the spike phases unchanged. This manipulation disrupts spike co-occurrence while preserving theta phase locking, allowing us to test whether theta phase locking alone can explain the population synchrony. We found that theta-cycle randomization significantly reduced the rate of synchronous events by 4.5 folds (Supplementary Figure 4). This new analysis demonstrates that theta phase locking alone cannot account for the population synchrony observed in our data.

Result (line 358)

“Correlated intracellular theta and theta-phase locking of the synchronous ensembles raise the question of whether population synchrony among CA1PCs extends beyond synchrony derived from these effects. To address this, we analyzed population synchrony after randomizing the theta cycles during which neurons spiked, while keeping their theta phases unchanged. Supplementary Figure 4 illustrates a significant reduction in synchronous event rates following theta cycle randomization. The finding indicates spiking at specific theta cycles plays a major role in driving population synchrony.”

(2) The definition of "synchrony" in the present work is very loose and refers to timescales of 20-30 ms. In previous literature that relates to synchrony of point processes, the timescales discussed are 1-2 ms, and longer timescales are referred to as the "baseline" which is actually removed (using smoothing, jittering, etc.).

Regarding the timescale of synchronous ensembles, we acknowledge that it varies considerably across studies and cell types. However, it is important to note that a timescale of dozens or even hundreds of milliseconds is commonly used in the context of synchrony terminology for CA1 pyramidal neurons(6,31-33). In fact, a timescale of 20-30 ms is considered particularly important for information transmission and storage in CA1, as it aligns with the membrane time constant of pyramidal neurons, the period of hippocampal gamma oscillations, and the time window for synaptic plasticity. Therefore, we believe this timescale is highly relevant and consistent with established practices in the field.

Reviewer #3 (Recommendations For The Authors):

(1) L19-20: "these synchronous ensembles were not associated with ripple oscillations" - this is a main fallacy in the present work (ripples are from the other side; there are not enough ripples to obtain sufficient statistical power to even test the hypothesis; etc.). The sentence should be removed.

As we have addressed in the public review, most ripples occur synchronously in both hemispheres(3,26). Many studies have used contralateral LFP to infer ripple occurrence on the ipsilateral side(26-29). Moreover, our new data now support the dissociation between synchronous ensembles and ripples with a much larger number of ripples and rigorous statistical testing (Figure 3-figure supplement 3 of the manuscript). These findings support our conclusion that synchronous ensembles are not associated with ripple oscillations.

Result (line 266)

“In total, 1052 synchronous ensembles and 174 ripple episodes were detected across these imaging sessions. Consistent with findings from walking animals, few synchronous ensembles occurred during ripples when animals were immobilized in a tube (Figure 3-figure supplement 3A, B). Moreover, no distinguishable ripple oscillations were observed in synchronous events, and the average firing rates during ripple episodes were near zero (Figure 3-figure supplement 3C, D). At the single-cell level, 90% of neurons showed significant negative spiking modulation during ripples, with ripple modulation indexes close to -1, indicating strong suppression of spiking (Figure 4Ai). This suppression extended to subthreshold membrane potentials, as nearly all cells exhibited decreased fluorescence

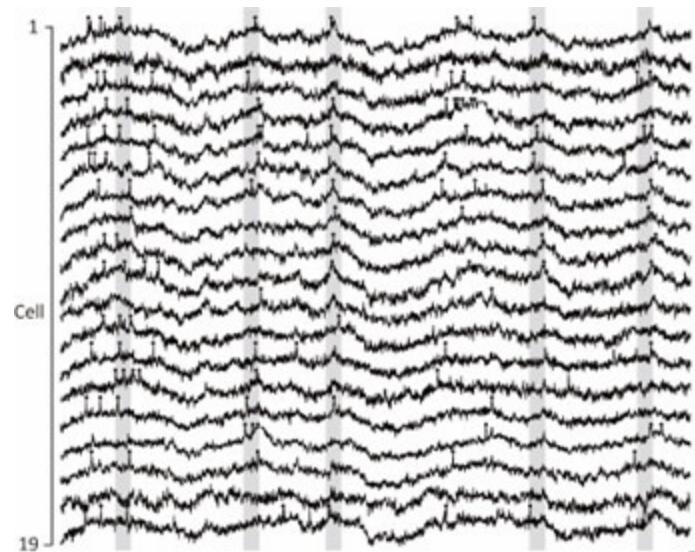
during ripples compared to baseline (Figure 4Bi, Ci). These results demonstrate that spiking activity and subthreshold membrane potentials are robustly suppressed during ripples.”

(2) L135/Figure 1: panel C and elsewhere: show the same traces after removing (clipping) the spikes. You may be able to see the intracellular theta nicely, which may be very strongly synchronized between neurons and could then be supplemented by ticks (as in conventional raster plots). This will allow a clearer visualization of the spiking and their relations with Vm.

We have created the plot as suggested (Author response image 2). As demonstrated in our figures (Figure 5 in the manuscript), the subthreshold membrane potentials of individual neurons are strongly correlated and coherent at theta frequency, consistent with the reviewer’s viewpoint.

Author response image 2.

Fluorescence traces of 19 simultaneously recorded cells with truncated spikes replaced by dots. Horizontal scale bar: 25 ms; vertical scale bar: -3%.



(3) Related to the above comment, in general, a much more robust approach with the present dataset may be to derive an estimate of the LFP from the intracellular records. Extracellular theta is related to intracellular theta (approximately the negative), and extracellular ripples co-occur with intracellular high-frequency oscillations. However, because the precise transfer function (TF) between the two is not well established, ground truth data should first be collected. This may be done by voltage imaging of even a single neuron in parallel with an extracellular glass pipette placed in near proximity of the same cell, at the same depth. Such datasets have been collected in the past, so it may be sufficient to contact those authors and derive the TF from existing data. Alternatively, new experiments may be required. It is possible that the TF will not be well defined - in which case there are two options: (1) limit the analyses to the relation between spikes in Vm, or (2) record new datasets with true LOCAL field potentials in every case.

We thank the reviewer for the insightful suggestion. Establishing a precise TF between intracellular and extracellular recordings is indeed crucial when exact phase information is required to draw conclusions. However, our goal is to understand the occurrence of specific network oscillation states surrounding these synchronous ensembles, rather than

pinpointing the precise phase at which they occur. Therefore, we believe that the strong bilateral cooccurrence of both theta and ripple oscillations provides a practical and valid foundation for supporting our objective.

While the approach suggested by the reviewer is an excellent idea, conducting simultaneous voltage imaging and local LFP recording is currently not feasible due to technical constraints associated with the implanted glass windows. Nevertheless, we recognize the potential value of this approach and plan to incorporate it into future experimental designs, which could provide further insights into the specific oscillatory phases associated with population synchrony.

(4) L135/Figure 1: panel D and elsewhere: Account for second-order spike train statistics (e.g., bursts). The simplest way to do this is to remove all spikes that are not the first spike in a burst. Otherwise, the zero-lag bin of a pair-wise CCG will be filled with counts that are due e.g., to the first spike of the second neuron co-occurring with the last spike in a burst of the first neuron. In other words, without accounting for bursts, sequential activity may be interpreted as synchrony.

We thank the reviewer for this insightful comment. As recommended, we have performed the suggested analysis by removing all spikes that are not the first spike in a burst (Supplementary Figure 3). The results demonstrate that, even after removing the subsequent spikes in bursts, the rates of synchronous events remain unchanged compared to the original data, and the sizes of the synchronous ensembles are also unaffected. These findings indicate that our conclusions are robust and not confounded by the presence of later spikes within bursts.

Result (line131)

“The observed population synchrony was not attributable to spikes in complex bursts, as synchronous event rates did not differ significantly with or without the inclusion of later spikes in bursts (Supplementary Figure 3).”

(5) L135/Figure 1: panel D and elsewhere: Related to the previous comment: the "grand average" CCG of a single neuron with all the other simultaneously recorded neurons is prone to a peak at zero lag ("synchrony") even if all pairs of neurons have pure mono-synaptic connections (e.g., at a 2 ms time lag). This is because neuron1 (N1) may precede N2, whereas then N3 may precede N2. In such a case, the pooled CCG will have two peaks - at e.g., 2 ms and -2 ms. However, if bursts occur (as is the case in CA1 and Figure 1C), there will also be non-zero counts around zero lag, which will accumulate as well. Together, these will build up to a peak around zero - even without any theta phase locking or any other alternative correlations.

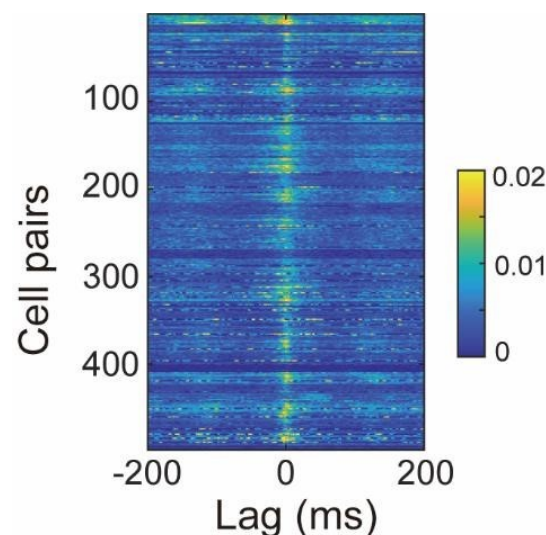
Please see our reply to comment #6 below.

(6) L135/Figure 1: panel D and elsewhere: refrain from averaging "grand averages" over neurons. This problem is distinct from the above (where e.g., N2-N1 is averaged with N2-N3). In any case, all visualizations and measures should be derived from individual (pair-wise) CCGs, and not "grand averages"

We thank the reviewer for the detailed comments and appreciate the opportunity to clarify our methods and analyses related to population synchrony. In response to the suggestion to replace grand average CCGs with pairwise CCGs, we have now included a heatmap to visualize individual pairwise CCGs for all recorded neuronal pairs that meet our inclusion criteria (497 pairs, Author response image 3). The heatmap provides a comprehensive view of the temporal relationships between neuron pairs.

Author response image 3.

Color-coded plot of pairwise CCGs for all cell pairs that meet our inclusion criteria.



While we have chosen to keep the grand-average CCGs, we emphasize that they are served only to summarize the overall temporal scale of the population synchrony. Importantly, our conclusions regarding synchronous ensembles are not based on grand-average CCGs. Instead, we assess population synchrony using a rigorous approach: we compute spike counts across the population in 25-ms sliding windows and compare these counts to those derived from jittered data, where spike timings are randomly shifted by ± 75 ms while preserving the overall spike count distribution. Synchrony is identified when the original spike counts exceed those from the jittered data by more than 4 standard deviations. This approach accounts for the potential accumulation of zero-lag counts arising from mixed mono-synaptic connections or bursting, as noted by the reviewer. By perturbing spike timings and preserving spike count distributions, our method identifies synchrony beyond what is expected by chance, ensuring robust and artifact-free conclusions.

(7) L135/Figure 1: panel D and elsewhere: after deriving measures (peak lag, FWHM, synchrony strength, etc.) from individual pairwise CCGs, show the measures as a function of the spike counts. For a pair of neurons N1-N2, derive the geometric mean spike count (or the mean, or the max). For instance, if there are 500 pairs of neurons, show e.g., pairwise synchrony strength as a function of the spike count geometric mean. While little correlation is expected when the timescale is small (1-2 ms), the "synchrony" effect at a timescale of 20-30 ms is expected to be very strongly related to the spike counts. Because the spike counts may differ between the lower and higher speed "states", many results reported in the present manuscript may be an epiphenomenon of that relationship.

We thank the reviewer for these valuable comments. In response, we analyzed pairwise synchronization strengths as a function of spike counts geometric mean of neuron pairs, as suggested. As shown in Author response image 4, the CCG peak counts in the original data (red dots) increase with the spike count geometric mean, consistent with the expected trend. However, this trend is also captured by the jitter control (black dots), which reflects synchrony levels expected by chance given the spike count levels.

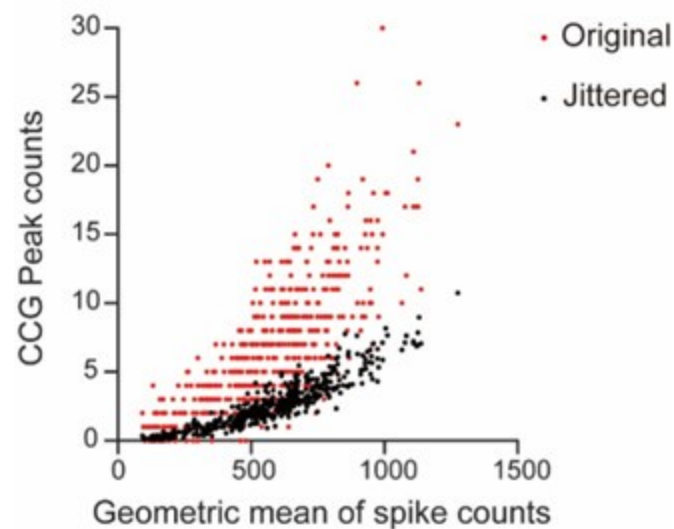
Importantly, the normalized synchronization strengths - defined as the ratio of CCG peak counts in the original data to the jitter control - are not positively correlated with spike

counts and remain significantly greater than 1 across all spike count levels (Author response image 5). This demonstrates synchrony beyond what could be explained by spike count variations alone.

While we understand the potential influence of state-dependent spike count variations, our jittering approach effectively controls for this by removing chance-level synchrony that could arise from these variations. This ensures that the observed synchrony reflects genuine neuronal interactions rather than an epiphenomenon of spike count variations between states.

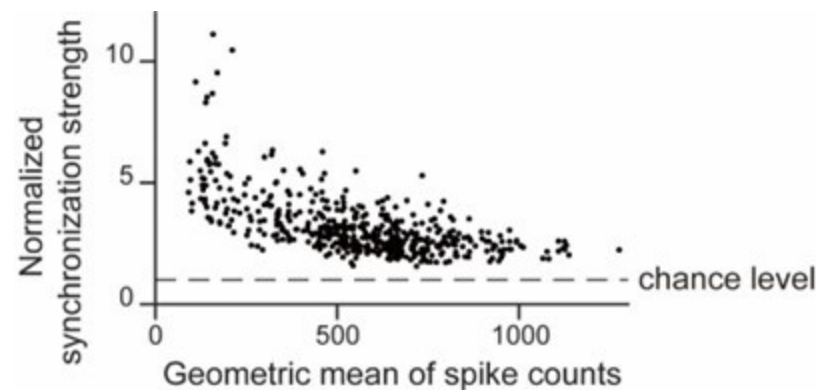
Author response image 4.

Plot of peak spike counts of pairwise CCGs (red) and mean spike counts from jittered data (black) against geometric means of pair spike counts.



Author response image 5.

Plot of normalized synchronization strengths against spike count geometric means.



(8) L135/Figure 1: show all CCGs in a color matrix.

We have generated a color matrix visualization of all pairwise CCGs, as recommended (Author response image 3). This visualization highlights the consistency of our results across

neuron pairs.

(9) L168/Figure 2: the LFPO is nearly irrelevant - it is from the other hemisphere, and it is unclear whether the depth is the same as in the "deep" (closer to the brain surface) imaging plane used for the voltage recordings.

As previously explained, the LFPO is relevant because it reveals the occurrence of theta and ripple states, which are highly synchronous across both hemispheres and serve as reliable indicators of network states relevant to our findings.

(10) L222/Figure 3: The ripple-related analyses are completely irrelevant - ripples are a local phenomenon, and recording from the other hemisphere is completely irrelevant.

We thank the reviewer's suggestions. As we have explained in the public review, as well as in the reviewer's comments #1 and #3, the occurrences of theta and ripple oscillations are well-coordinated across hemispheres. As our analyses only depend on the occurrences of these oscillations, our conclusions regarding the association of the synchronous ensembles with theta but not ripple oscillations are supported by data.

(11) L292/Figure 4, panels A-E: please trigger Vm on the same-neuron spikes, not on the "synchrony events". This will already explain most of the observations. Some of this is already shown in the supplementary figures.

As the reviewer correctly noted, we have already presented data triggered on same-neuron spikes in Figure 5-figure supplement 1C and D. The reason we show synchrony-triggered LFP and subthreshold Vm in the figure is to highlight the network dynamics during synchronous events. This approach provides a broader perspective on how neural networks function and interact during periods of synchrony, offering insights beyond individual neuron activity

(12) L351/Figure 5, panel C: typo - should read "strength"

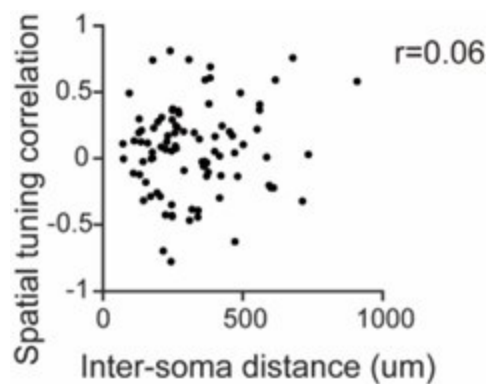
The typo has been corrected.

(13) L351/Figure 5: show "spatial tuning correlation" vs. inter-soma distance (as in Fig. 4G). This may explain part (if not all) of the observations

We have followed the reviewer's suggestion and generated the plot (Author response image 6). Consistent with the literature, the plot demonstrates that the spatial tuning correlations of place cell pairs exhibit little relationship with their inter-soma distances.

Author response image 6.

Plot of spatial tuning correlation vs. inter-soma distance (Spearman correlation coefficient=0.06, $p=0.54$, $n=91$ pairs).



(14) L937/Figure S3: panel A: the ripples here appear to be recorded from the top part of the layer, i.e., the electrode is not in the center of the layer. Panel B: add statistical testing.

We agree with the reviewer that this is possible, as we aimed to place our LFP electrodes in the *stratum pyramidale*. Regarding panel B of the figure, we verified the quality of LFP recordings by acquiring data from subsequent sessions following the initial imaging sessions. The detection of ripples in the same animals during these later sessions indicates that the absence of ripples during the first sessions is not due to deterioration in LFP recording quality. However, due to the small sample size, the statistical power is insufficient to demonstrate significance ($n=5$ sessions, $p=0.06$, Wilcoxon signed-rank test). Nevertheless, our conclusions are not contingent upon achieving statistical significance in this test.

(15) L944/Figure S4: The " $R=1$ " is very likely to be an outcome of $n=1$ spike. In other words, estimates of phase are unreliable when the spike count is very low. This is related to the problem referred to in Comment #7 above.

We understand that phase estimates can be unreliable when the spike counts are low. We now highlight that this effect has been taken into account by a shuffling procedure that assesses the significance of phase modulation, and by excluding neurons with nonsignificant modulation strengths. Neurons with low spike count or inconsistent spike phases are typically excluded due to the non-significant strength of phase modulation.

Method (line 828)

"The significance of the modulation strength was tested by shuffling the spike timings and recalculating the modulation strength a thousand times to generate a distribution based on the shuffled spike timings. The original modulation strength was then compared to the distribution, with significance determined if it exceeded the 95% confidence interval of the shuffled values.

Significant modulation strengths were plotted and compared across groups."

(16) L944/Figure S4: Putting the spike count issue (Comment #15) aside for a moment, the analyses in this figure are actually valid - they are carried out at the single-neuron level, with respect to the local (same-neuron) Vm. These findings provide a key alternative explanation to the observations purported in the main figures: (1) if spiking is locked to intracellular theta (occurring at the peak of Vm); and if (2) intra-cellular (Vm) theta is locked to extracellular theta (antiphase); and if (3) extracellular theta is similar for nearby neurons (the imaged neurons), then synchrony is a necessary outcome. The key question

is then whether there is any EXTRA synchrony between the CA1PC - beyond that which necessarily derives from (1)+(2)+(3).

We acknowledge the reviewer's perspective. However, the factors (1)+(2)+(3) alone do not account for the synchrony we observed. As the reviewer points out (and as discussed in our response to the public review and in Supplementary Figure 4), theta phase locking does not necessarily imply population synchrony. To demonstrate that population synchrony extends beyond the contribution of (1)+(2)+(3), we performed an analysis where the theta cycles in which neurons spike were randomized, while the theta phases remained unchanged (Supplementary Figure 4). The analysis revealed that randomizing the theta cycles while preserving theta phases significantly reduces population synchrony. This finding indicates that spiking in specific theta cycles plays a major role in driving population synchrony.

Result (line 358)

“Correlated intracellular theta and theta-phase locking of the synchronous ensembles raise the question of whether population synchrony among CA1PCs extends beyond synchrony derived from these effects. To address this, we analyzed population synchrony after randomizing the theta cycles during which neurons spiked, while keeping their theta phases unchanged. Supplementary Figure 4 illustrates a significant reduction in synchronous event rates following theta cycle randomization. The finding indicates spiking at specific theta cycles plays a major role in driving population synchrony.”

(17) L944/Fig. S4: Why 71 neurons in AB and only 59 in CD?

In the previous version, panels A and B included 71 neurons, as we collected data from 71 cells across 5 mice (see the text below).

Result (line 93)

“...in total, 71 cells imaged from 5 fields of view in 5 mice; Figure 1B and Supplementary Figure 1A and 1B).”

In the current version, we only include neurons with significant modulation strengths, reducing the number of cells from 71 to 65 in panel A and from 71 to 54 in panel B.

Methods (line 828)

“The significance of the modulation strength was tested by shuffling the spike timings and recalculating the modulation strength a thousand times to generate a distribution based on the shuffled spike timings. The original modulation strength was then compared to the distribution, with significance determined if it exceeded the 95% confidence interval of the shuffled values. Significant modulation strengths were plotted and compared across groups.”

“Figure 5-figure supplement 1 Figure legend (line 1231)

Polar plot comparing subVm theta modulation between spikes participating in synchronous ensembles (sync spikes) and spikes not participating in synchronous ensembles (other spikes) during immobility. Each dot represents the averaged modulation of a cell. Cells with modulation strengths that are not significant are excluded in the plot and in the comparison.”

For panels C and D, we excluded neurons with four or fewer triggering events from the analysis, which reduced the number of cells from 71 to 59 (see the second text paragraph below).

Method (line 835)

“We extracted segments of fluorescence traces using a ± 300 ms time window centered on the spike timings. To examine variations in fluorescence waveforms triggered by spikes within and outside synchronous events, we categorized the fluorescence traces based on whether the spikes occurred within or outside these events. Subsequently, we performed pairwise comparisons of the fluorescence values from the same neuron, concentrating on spikes occurring during corresponding behavioral states. Neurons with four or fewer triggering events in any of these categories were omitted from the analysis.”

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